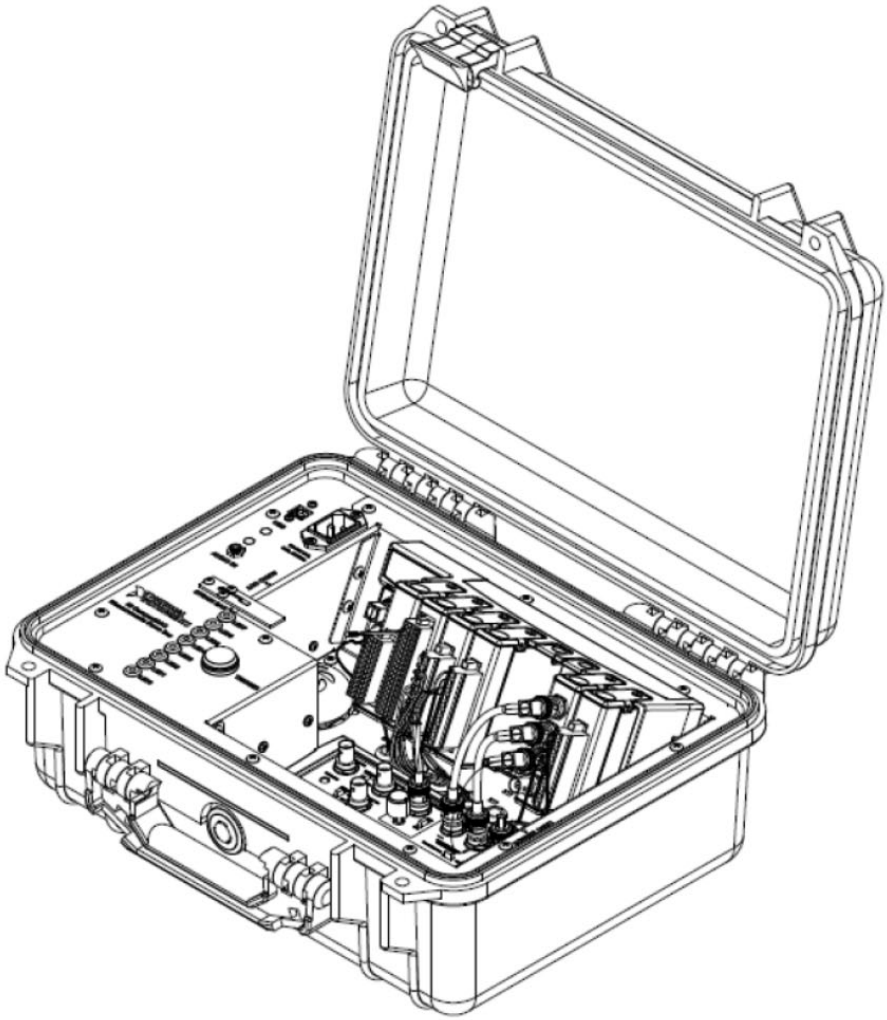
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HANDS-ON MANUAL

CompactDAQ Demo Kit

Learn how to take measurements in minutes, trigger a signal based on your data, and save to a structured file format.

ni.com/CompactDAQ

CompactDAQ and LabVIEW Hands-On Table of Contents

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Exercise: Log Vibration Data with Alarm

Estimated time: 1.5 - 2 hours.

Goals

- Learn how to acquire data from a CompactDAQ device and trigger an alarm with LabVIEW.

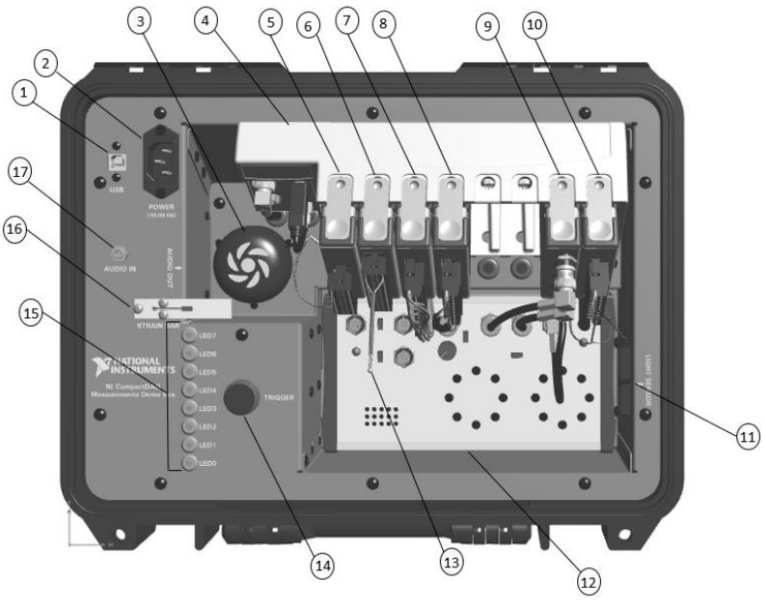
Equipment

Windows Laptop:

- Software: LabVIEW 2023 Q3 Recommended, DIAdem 2024 Q2 Recommended
- Driver: NI-DAQmx 2023 Q3

Hardware:

- CompactDAQ demonstration kit



1. USB Input
2. Demo Kit Power
3. Speaker
4. NI cDAQ-9178
5. NI 9236
6. NI 9213
7. NI 9472
8. NI 9263
9. NI 9234
10. NI 9215
11. Solar Panel
12. DSA Demo Box
13. J-Type Thermocouple
14. Trigger Button
15. LEDs (0-7)
16. Strain Gauge Bar
17. Audio Input Port

Figure 1. CompactDAQ Demonstration Kit

Part A: Acquire Vibration Data using the DAQ Assistant

Estimated time: 20 minutes.

In Part A, you will use LabVIEW to quickly set up a program to acquire acceleration data from the fan in the demo box using the DAQ Assistant.

1. Open LabVIEW.
2. Create a new VI by selecting **File** > **New VI** in the pop-up window.

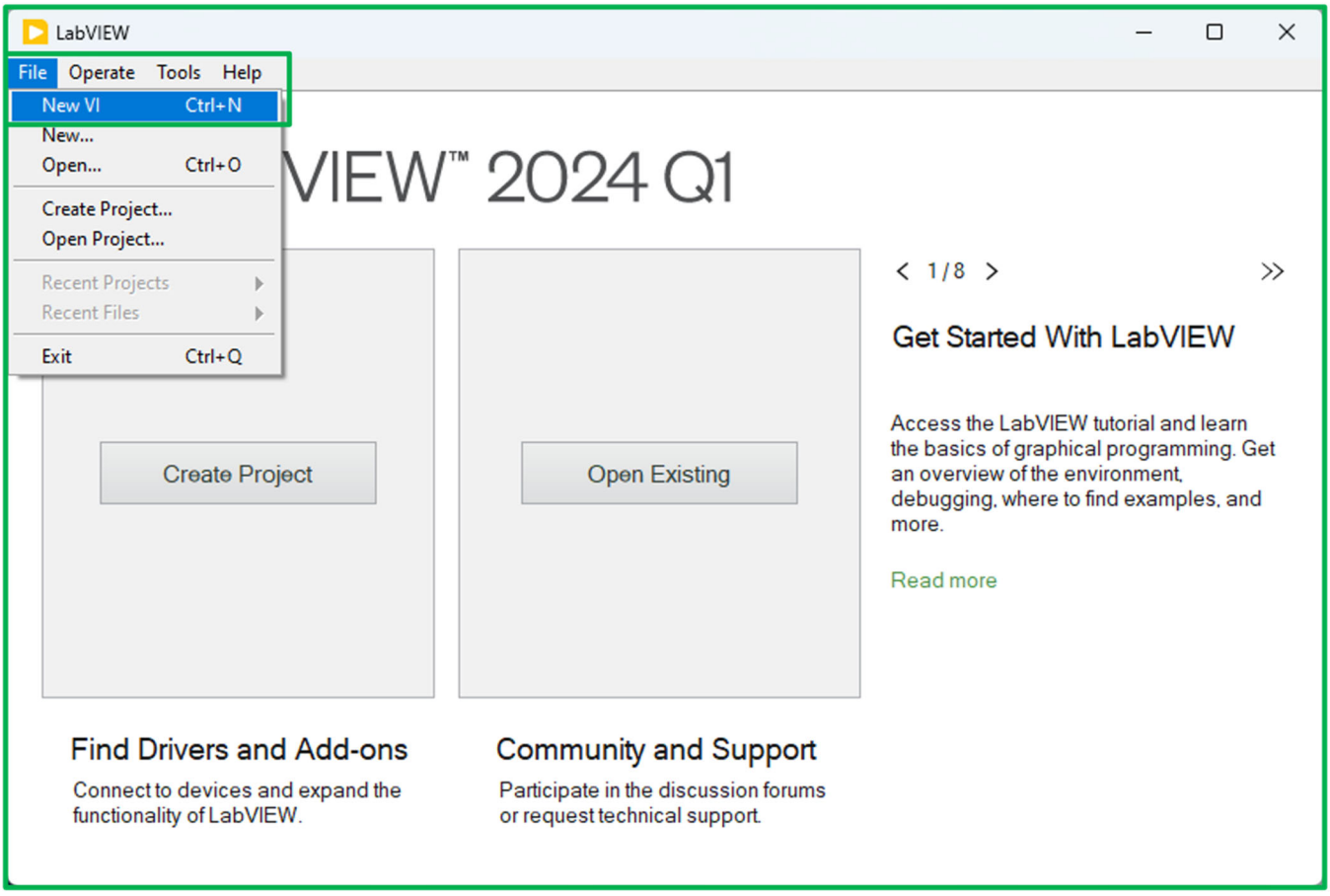


Figure 2. This step creates a new VI, which is made of the Front Panel that serves as the user interface, and the Block Diagram that contains the program code. Every VI will have both.

3. Once open, click **File** > **Save As** > **cDAQ+LV Exercise**
4. Navigate to the window with the Block Diagram by clicking **Window** > **Show Block Diagram** or using the keyboard shortcut CTRL + E.

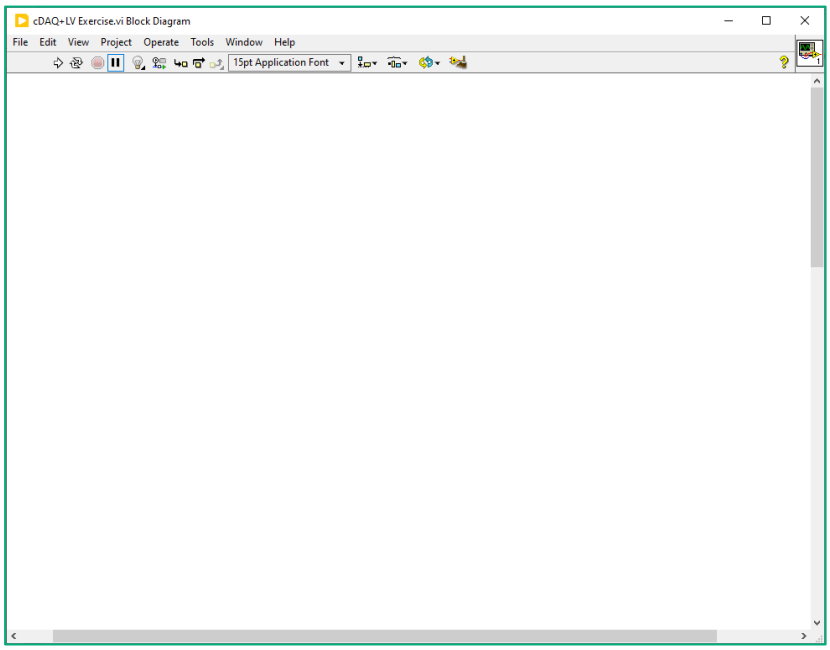


Figure 3. The Block Diagram is where you will add the graphical source code of your LabVIEW program. Block diagram objects include terminals, subVIs, functions, constants, structures, and wires, which transfer data among other block diagram objects.

5. To place a DAQ Assistant down, access the Quick-drop menu through the shortcut CTRL + Space.

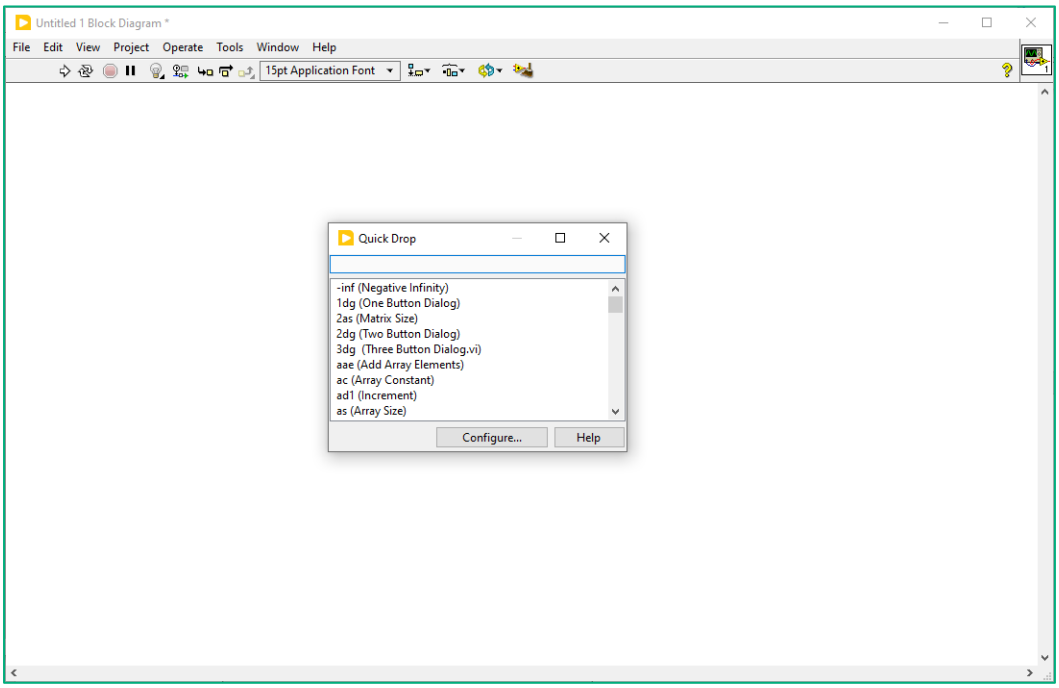


Figure 4. Quick Drop is a *quick* way to find functions and add them to your Block Diagram.

6. Type “DAQ” in the search field and double-click on the top suggestion labeled “DAQ Assistant”.

7. To place the DAQ Assistant down, click anywhere on the Block Diagram.

8. A **Create New** window will pop, and in this window, you will select the measurement type for the task in the DAQ Assistant.

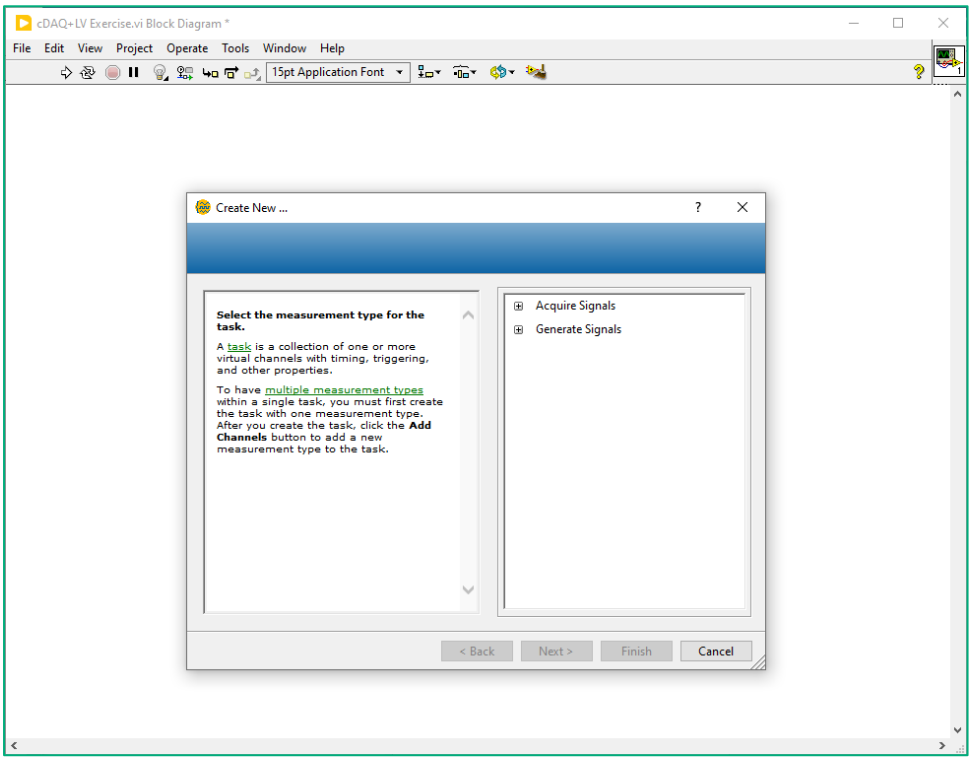


Figure 5. The DAQ Assistant is a configuration-based way to acquire data from physical or simulated NI DAQ hardware.

9. To configure an acceleration measurement, expand the **Acquire Signals** field, then the **Analog Input** field, and then the **Acceleration** field. Click **IEPE.**

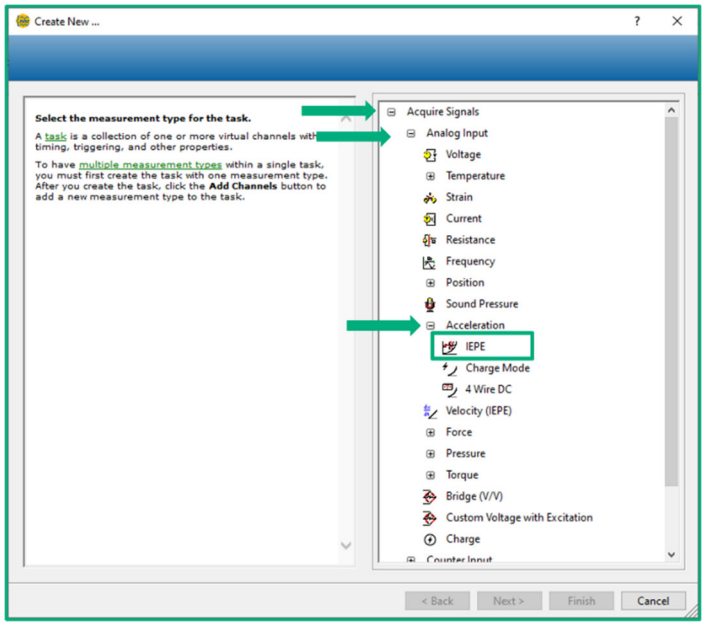


Figure 6. NI-DAQmx tasks are a collection of channels with homogenous timing and triggering properties.

10. Select channel **ai0** of **cDAQMod7 (NI 9234)** as the physical channel of the task by clicking it. Then click OK.

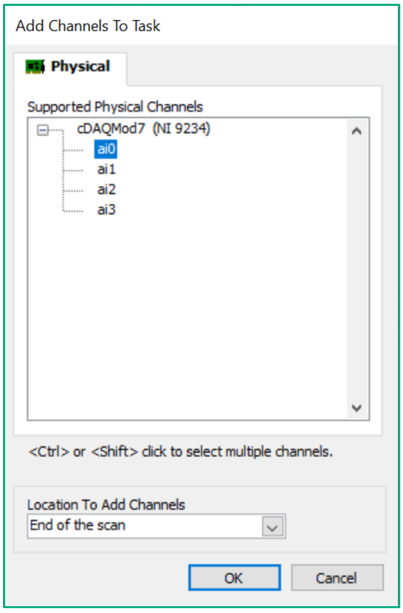


Figure 7. Channel ai0 measures the acceleration of the fan in the X direction.

11. In the **Timing Settings**, leave as the default. Click **OK** to confirm settings and close out of the DAQ Assistant window

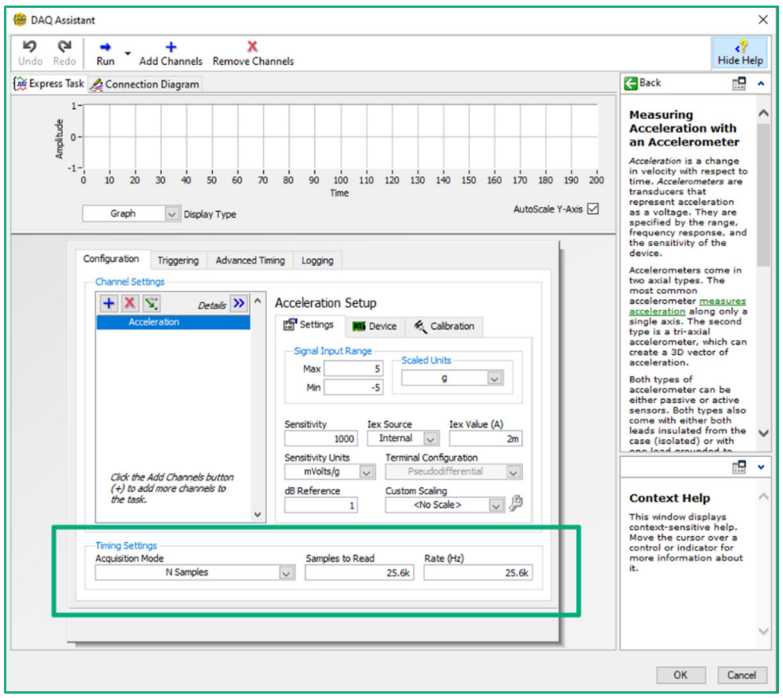


Figure 8. You can reopen this window to reconfigure your channel and timing settings at any time by double-clicking the DAQ Assistant on the block diagram.

12. Next you will create a way to visualize the raw acceleration data by creating a waveform graph indicator and wiring it to the output of the DAQ Assistant. Navigate to the front panel by using the short cut CTRL+E or use CTRL+T to view the Front Panel and Block Diagram side by side.

13. Right Click the Front Panel and navigate to **Fuse Design System** > **Graph** > **Waveform Graph**.

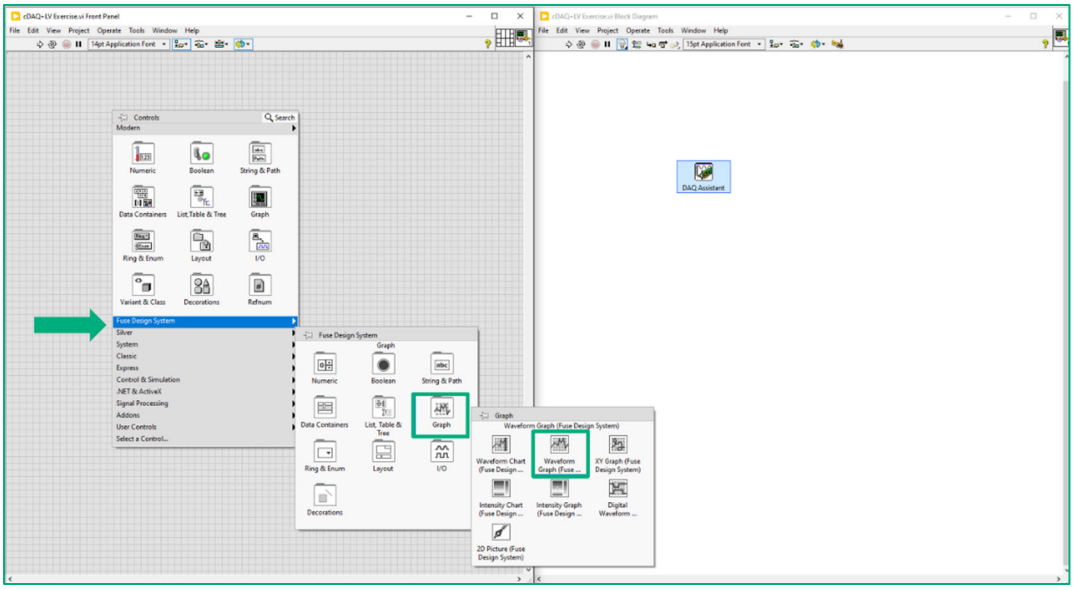


Figure 9. The Front Panel is the user interface and has controls and indicators, which are the interactive input and output terminals, respectively.

14. Click **Waveform Graph** and place it down by clicking anywhere on the Front Panel.

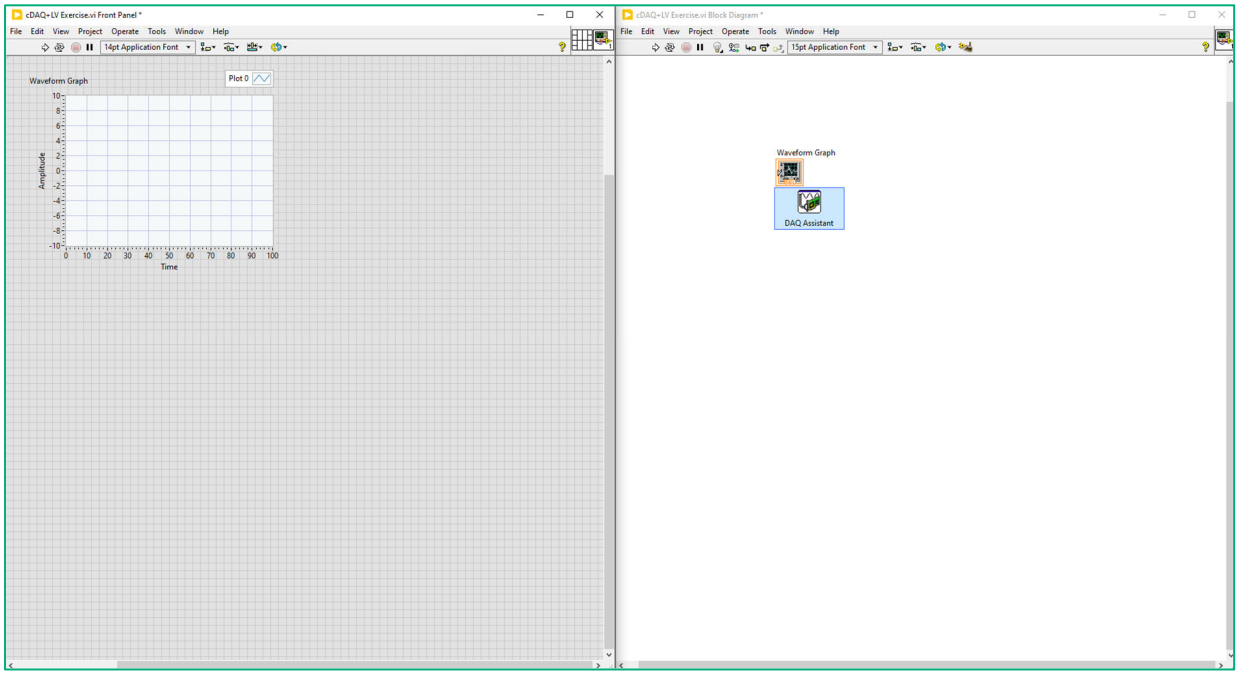


Figure 10. The Front Panel now includes a way to visualize your acceleration signal.

15. On the Block Diagram, click and drag the graph indicator to the right side of the DAQ Assistant, and wire the data from the DAQ Assistant to the indicator.

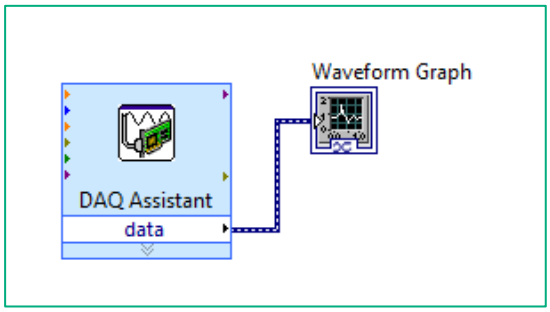
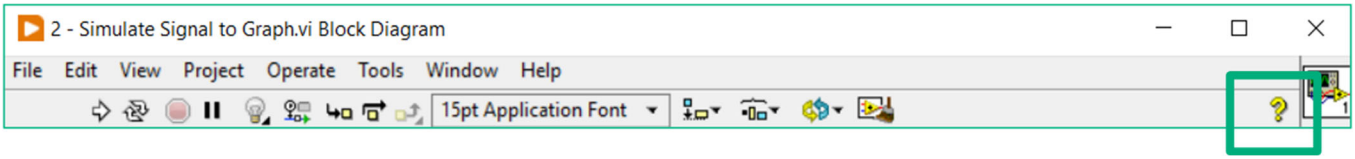


Figure 11. The DAQ Assistant passes data to the waveform graph via a wire on the Block Diagram.

LabVIEW Tip:



Press the **Context Help** button in the upper right portion of the Block Diagram and hover over any object to learn more about it.

16. Add a while loop around the DAQ assistant and graph indicator to collect data continuously by right clicking the Block Diagram and navigating to **Structures** > **While Loop**.

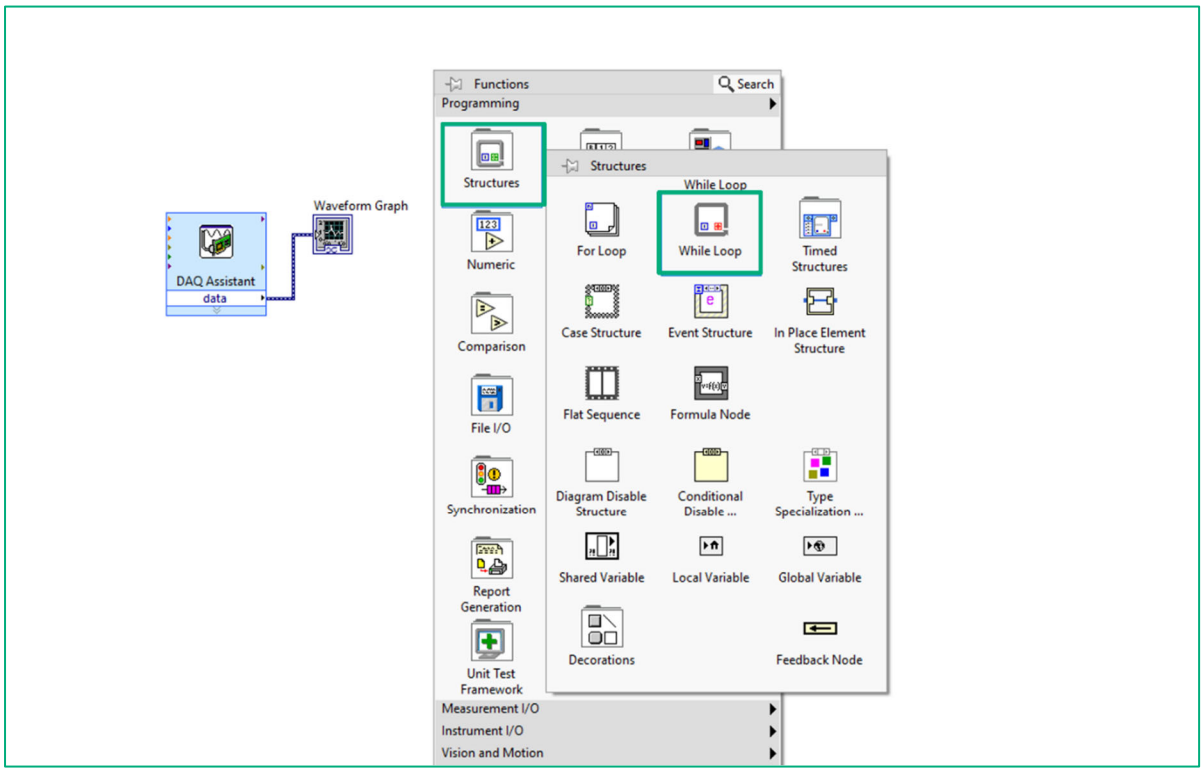


Figure 12. Without a while loop, LabVIEW will only collect one set of samples per run.

17. Click **While Loop** and draw a rectangle around the DAQ Assistant and waveform graph indicator.

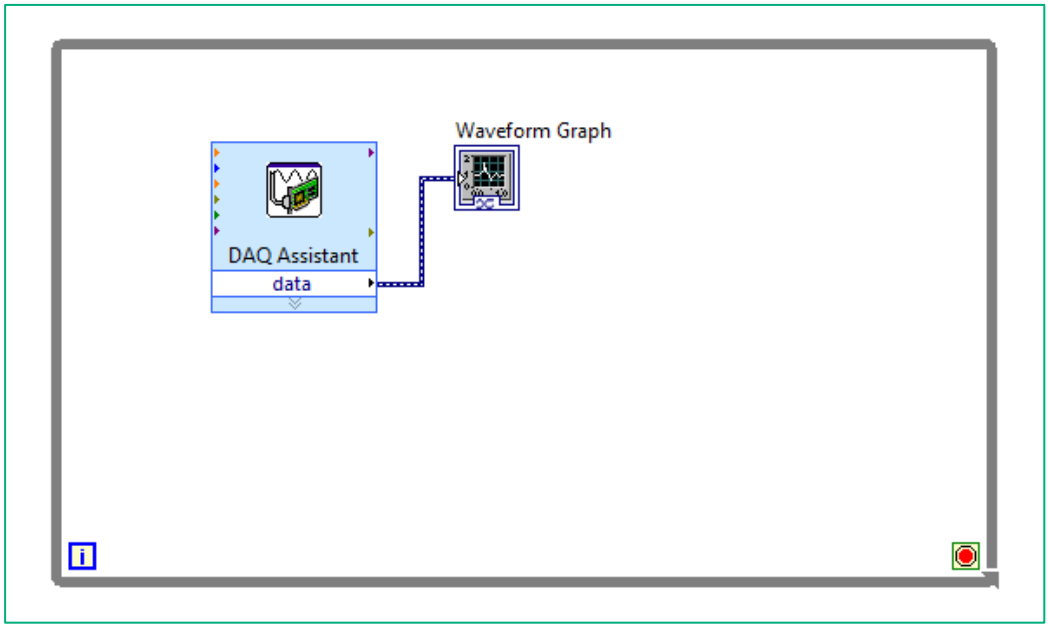


Figure 13. A While Loop repeats the code within its subdiagram until a specific condition occurs and always executes at least one time.

18. To stop collecting data at any point, add a boolean control on the Front Panel. Right click the Front Panel and navigate to **Boolean** > **Stop Button**.

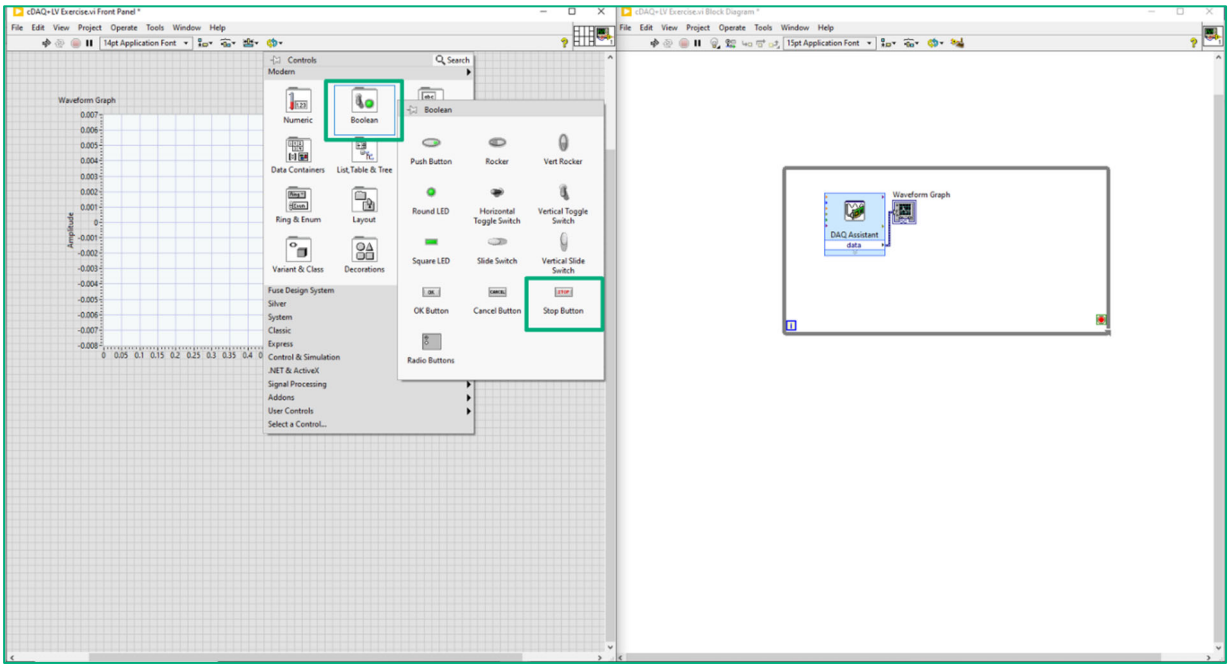


Figure 14. A boolean control can take many forms including a push button, toggle switch, LED, stop button, and more.

19. Click **Stop Button** and to place it, click anywhere on the Front Panel.

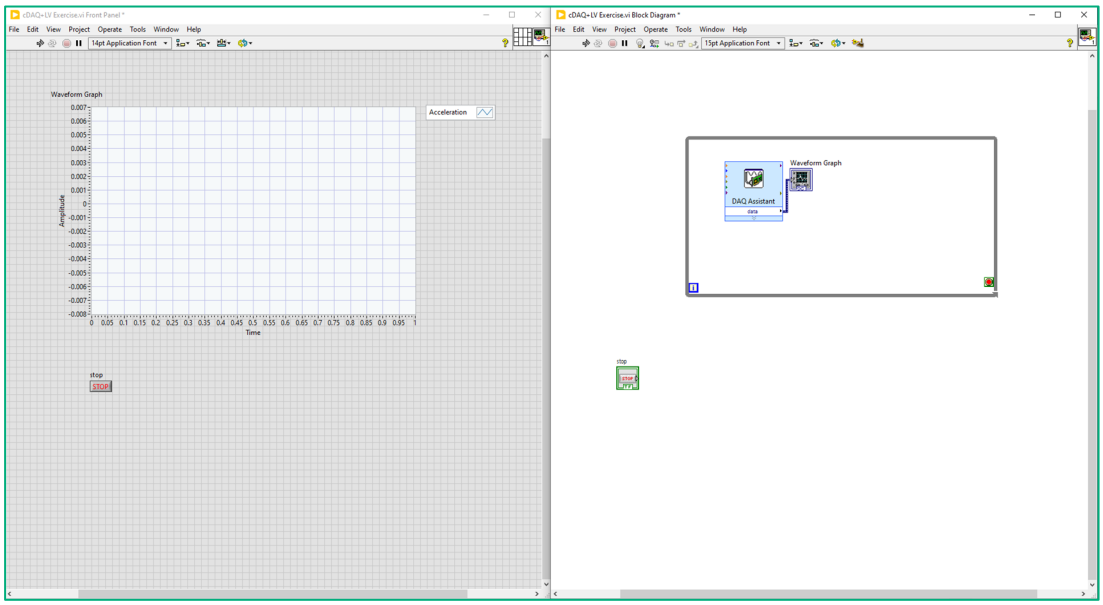


Figure 15. The stop button indicator shows up on both the Front Panel and Block Diagram, just like the waveform graph.

20. On the Block Diagram, click and drag the boolean control indicator inside the while loop and wire it to the loop control.

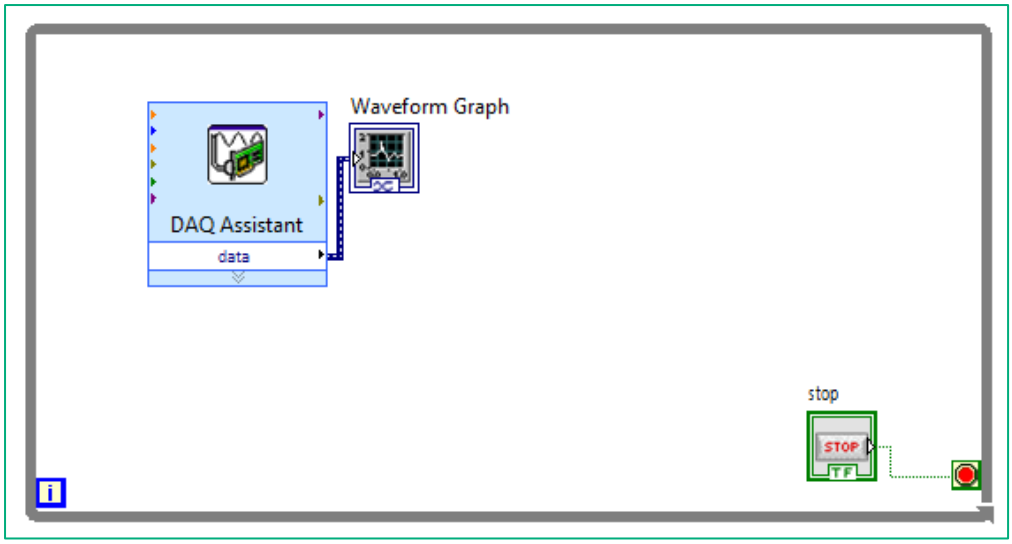


Figure 16. Once the stop button is wired to the loop control, clicking it on the Front Panel will stop the while loop.

21. To minimize the space taken up by the boolean indicator, right click it and click **View as Icon.**

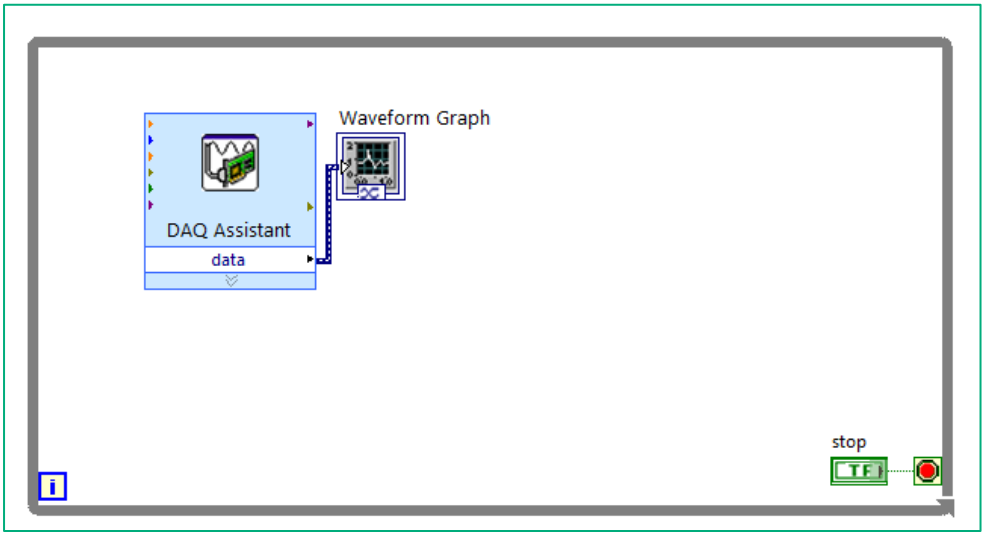


Figure 17. The Block Diagram for collecting vibration data is complete, and the VI can run.

22. To begin collecting data, Run the VI.



Figure 18. Click the arrow on the Front Panel toolbar to run the VI.

23. Turn the fan speed up or down with the knob and toggle the fan switch back and forth from **Balanced Fan** to **Unbalanced Fan**. Observe the effect of changing the fan conditions on the raw vibration (acceleration) data.

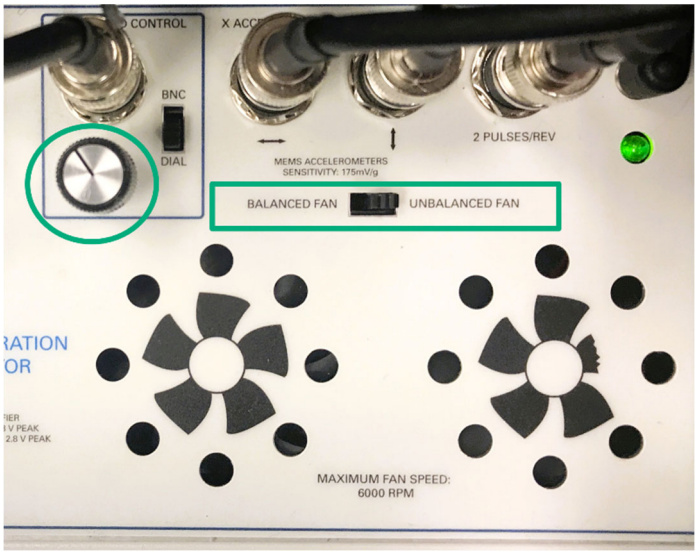


Figure 19. The fan controls on the vibration simulator box.

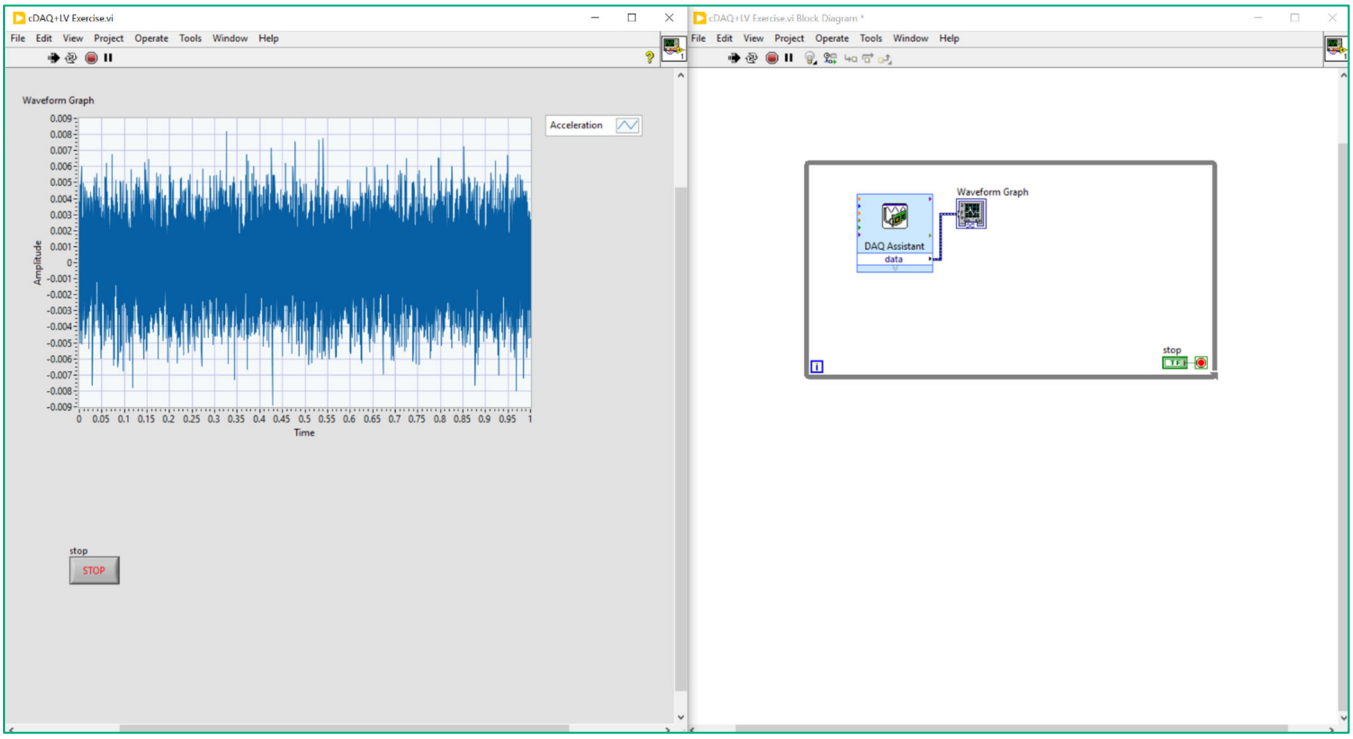


Figure 20. Example of the acceleration data on the Front Panel.

24. When finished, press the STOP button on the Front Panel.

Exercise Key Concepts:

In this exercise, you used the DAQ Assistant to configure an acceleration measurement, visualized the acceleration signal on a waveform graph, and used a while loop to collect data continuously. The DAQ Assistant is a helpful tool for taking measurements in minutes.

<End of Exercise>

Part B: Analyze the Signal

Estimated time: 20 minutes.

In Part B, you will add an analysis function to the program to compute the Root Mean Square (RMS) vibration of the signal. Each time the while loop iterates, the program collects 25.6k samples, and you want to calculate the RMS value for each set of samples.

Exercise Note:

A vibration measurement is a measurement of a dynamic quantity, which constantly changes. The RMS of the acceleration data is a key metric in vibration analysis because it is the representation of energy or the effective value of the signal.

In part C, you will compare the RMS values to a pre-set value and trigger an alarm if the RMS exceeds it at any point.

1. On the Block Diagram, access the Quick Drop menu again by pressing CTRL+Space, and in the search field type **RMS vi.** Double-Click the top suggestion and click to place it inside the while loop.

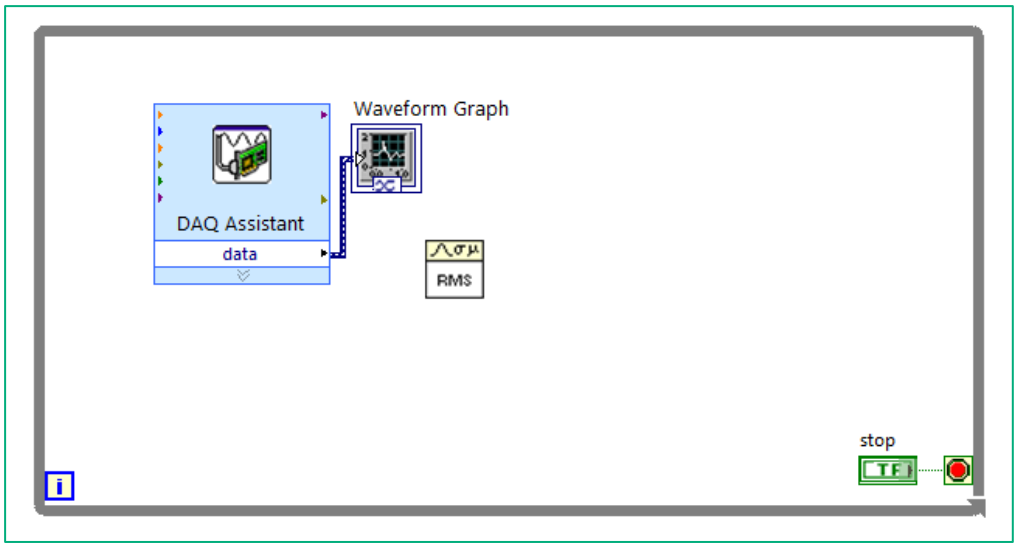


Figure 21. the RMS VI computes the root mean square (RMS) of the input sequence from the DAQ Assistant.

2. Next, wire the output data from the DAQ Assistant to the input terminal of the RMS VI.

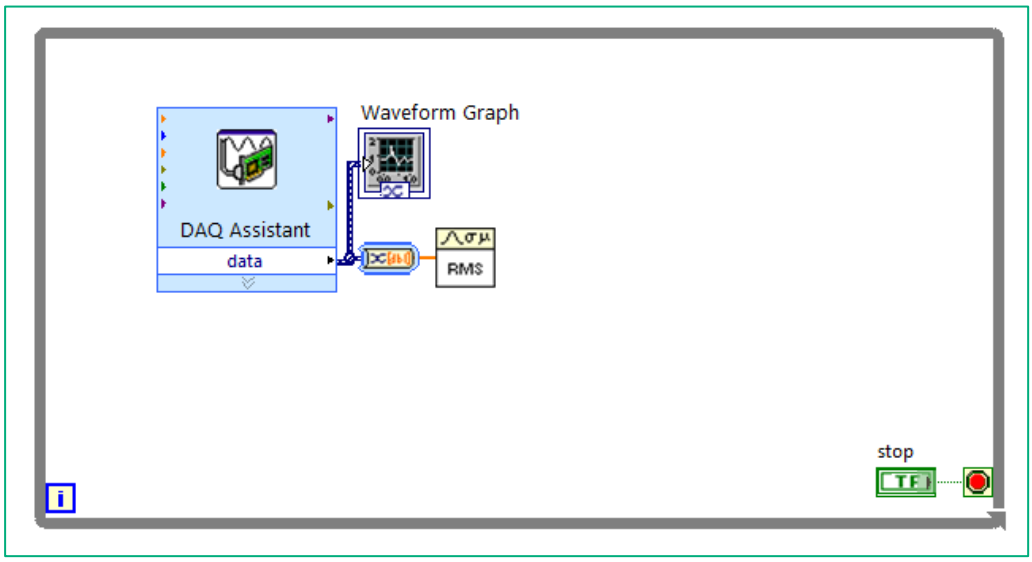


Figure 22. Every time the loop iterates and 25.6k samples are collected, the RMS VI will compute the RMS value for that dataset.

3. To collect the RMS values of all loop iterations, you need to use a shift register to pass the RMS value from one iteration to the next. First, wire the output of the RMS VI to the right-hand side of the while loop.

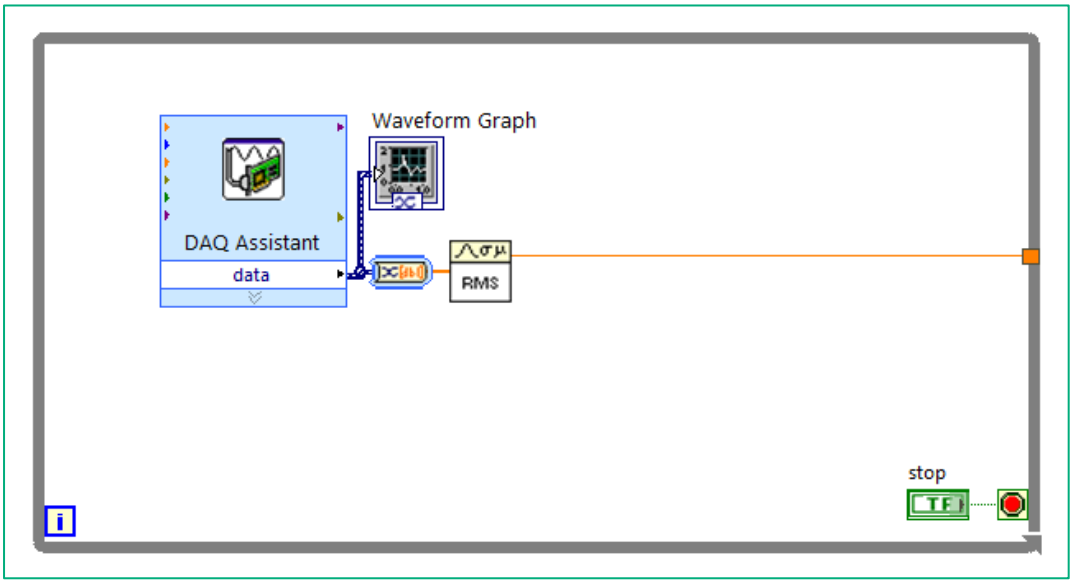


Figure 23. The orange square on the right side of the While Loop is called a "loop tunnel".

4. Right-click the loop tunnel (orange square) that you just made and click **Replace With Shift Register**.

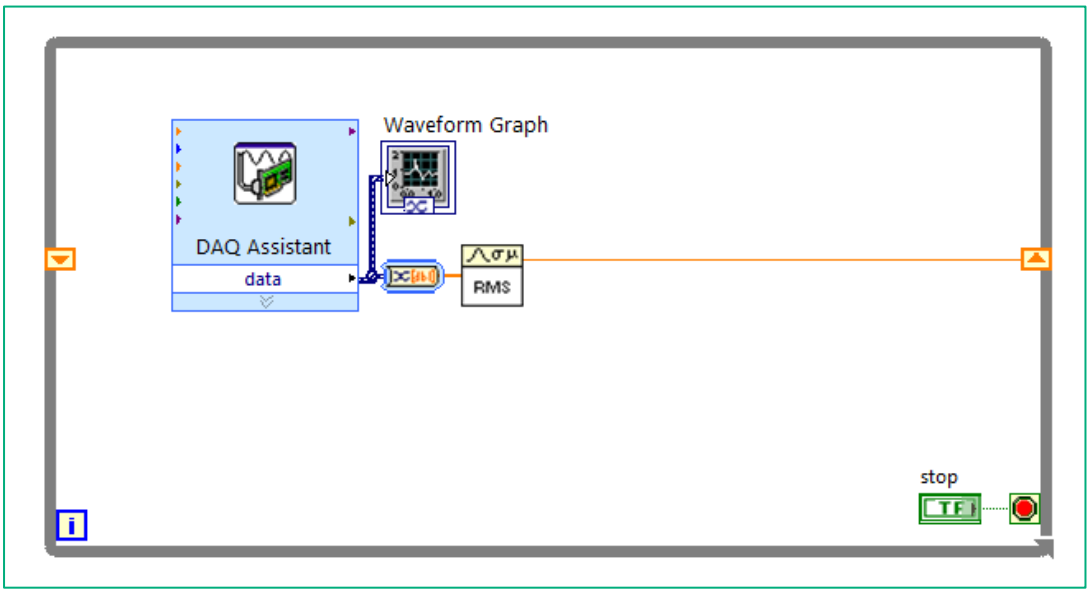


Figure 24. A shift register appears as a pair of terminals, directly opposite each other on the vertical sides of the loop border.

LabVIEW Concept:

Shift registers, which are similar to static variables in text-based programming languages, are used to pass values from one loop iteration to the next. After the loop code executes, data enters the right shift register and is passed to the left shift register on the next iteration of the loop.

5. To store the values from each iteration, you need an array.

6. Access the Quick Drop menu by pressing CTRL+ Space, and type Build Array.

7. Double-click the top suggestion and click inside the while loop, under the DAQ Assistant to place it. Then, wire the output of the left-hand shift register to the Build Array function.

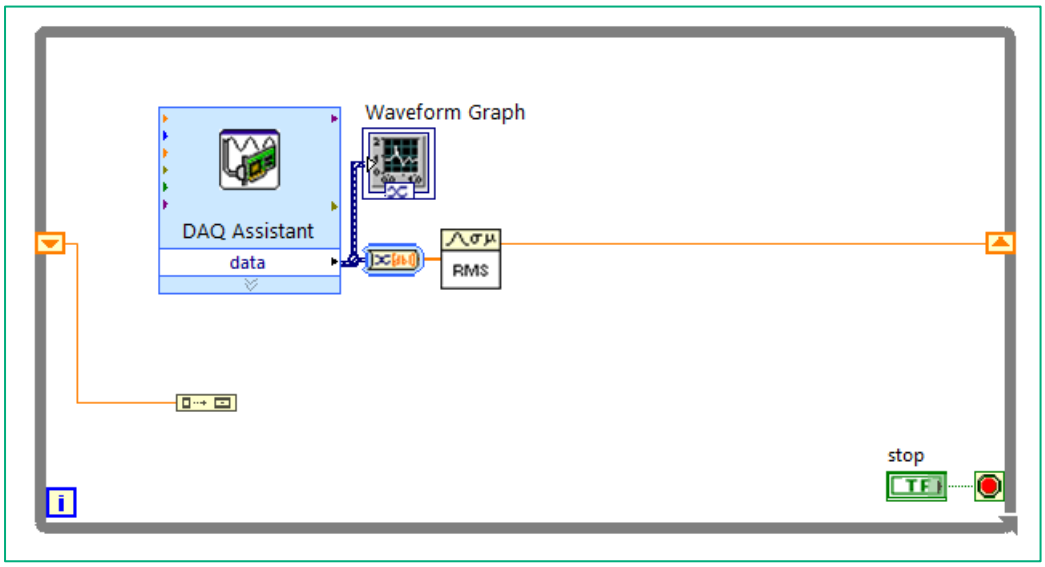


Figure 25. The Build Array function concatenates multiple arrays or appends elements to an n-dimensional array.

8. Now, every time the loop iterates and 25.6k samples are collected, the RMS VI will compute the RMS value and pass it to the build array function. This will continue for as long as the loop iterates. To observe the values, place a probe on the input data of the build array function by right clicking the wire and clicking **Probe.**

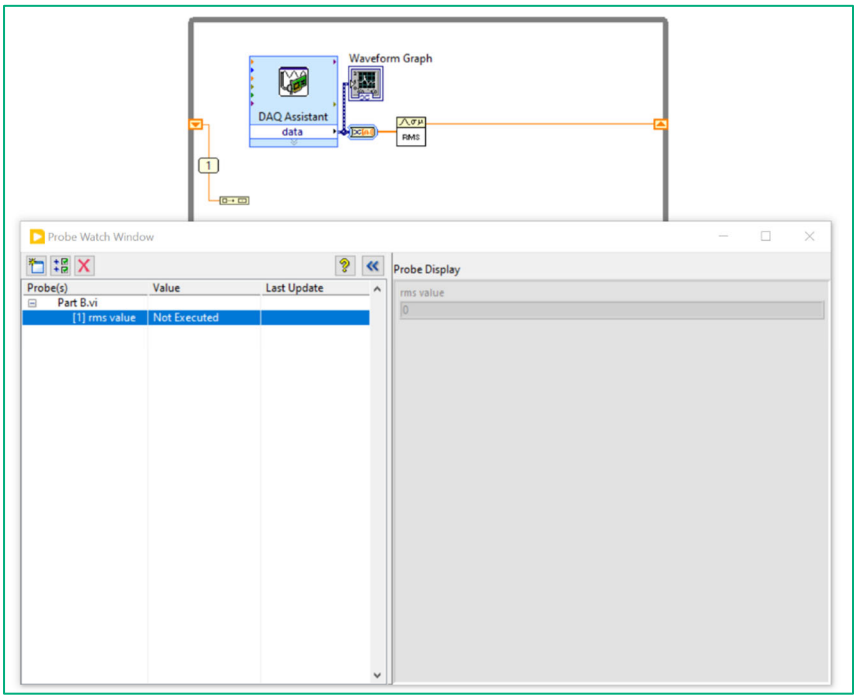


Figure 26. The Probe Watch Window will pop up to display the probe values.

9. Run the VI to start collecting data.

10. In the **Probe Watch Window**, observe the effect on the RMS data as you turn the fan speed up or down with the knob and toggle the fan switch back and forth from **Balanced Fan** to **Unbalanced Fan**.

11. Close out of the Probe Watch Window and click **Yes** on the pop up indicating that all probes will be closed.

12. When finished, press the STOP button.

Exercise Key Concepts:

In this exercise, you used a built-in LabVIEW function to calculate the RMS of the acceleration data for each loop iteration and passed the values to a **build array** function. You also practiced debugging skills when probing a wire to observe the RMS acceleration values as the VI ran. The **Probe** tool is also useful for comparing the data that appears in the **Probe** window to what you expected.

<End of Exercise>

Part C: Trigger an Alarm

Estimated time: 25 minutes.

In Part C, you will add a trigger such that an LED turns on when the RMS vibration exceeds a set value.

1. Place a comparison function on the Block Diagram by right clicking anywhere and navigating to **Comparison > Greater?** and clicking **Greater?** Click in the while loop to place it next to the RMS VI.

2. Wire the output data of the RMS VI to the top input of the comparison function.

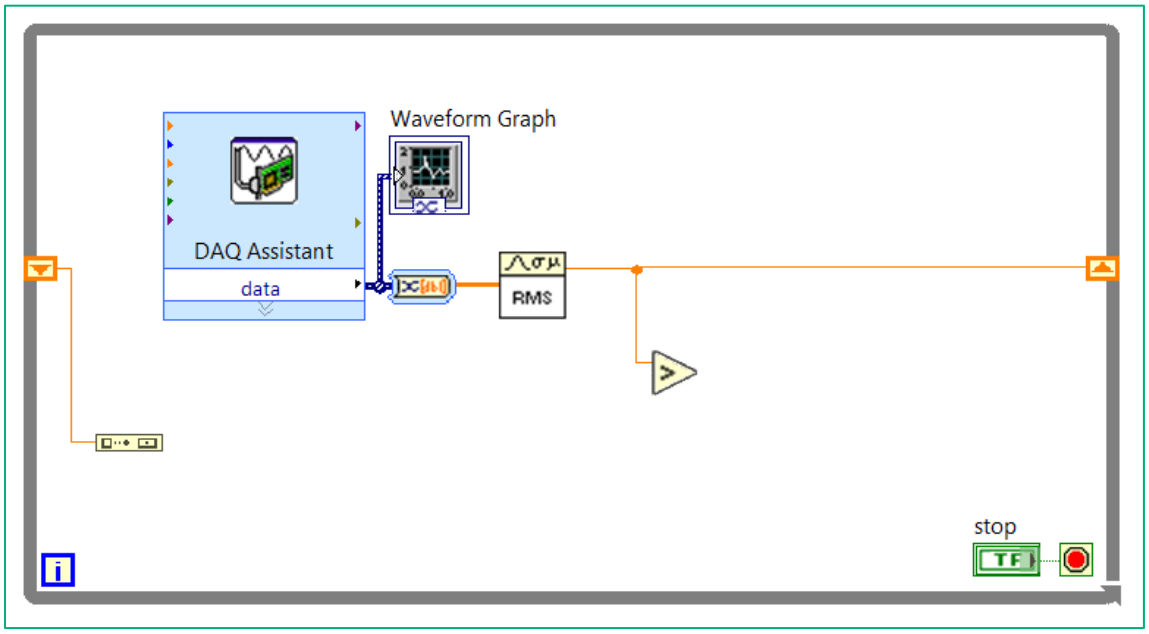


Figure 27. The comparison function returns a boolean value of TRUE or FALSE from the two inputs, once wired.

3. Next, to create a set point with which to compare the RMS value, right-click the other input of the comparison function and navigate to **Create > Control.**

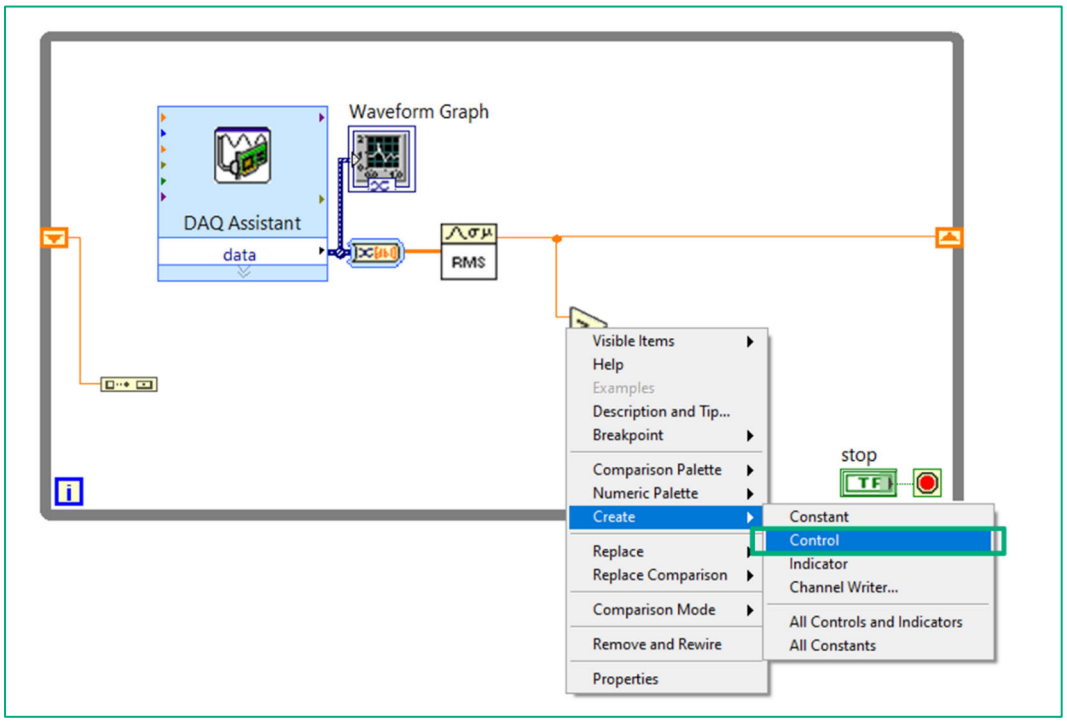


Figure 28. Creating a control rather than a constant allows you to change the input value from the Front Panel.

4. Click **Control** to place it on the block diagram and rename it to “Set Point”. It will be wired automatically to the comparison function.

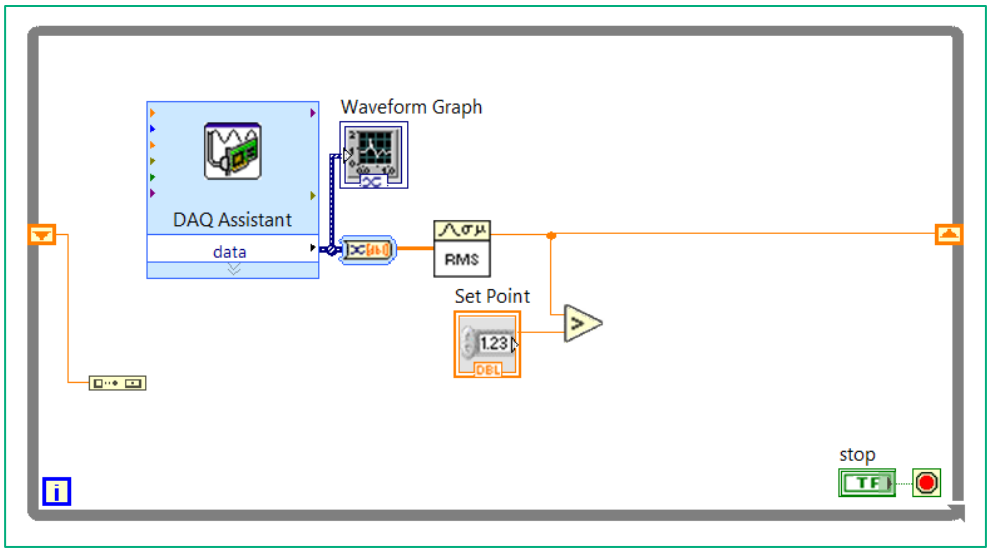


Figure 29. The RMS value of each loop iteration can now be compared with the set point value.

5. Next, to minimize the space that the control takes up in the while loop, right click it and click **View as Icon** just like you did in Part A.

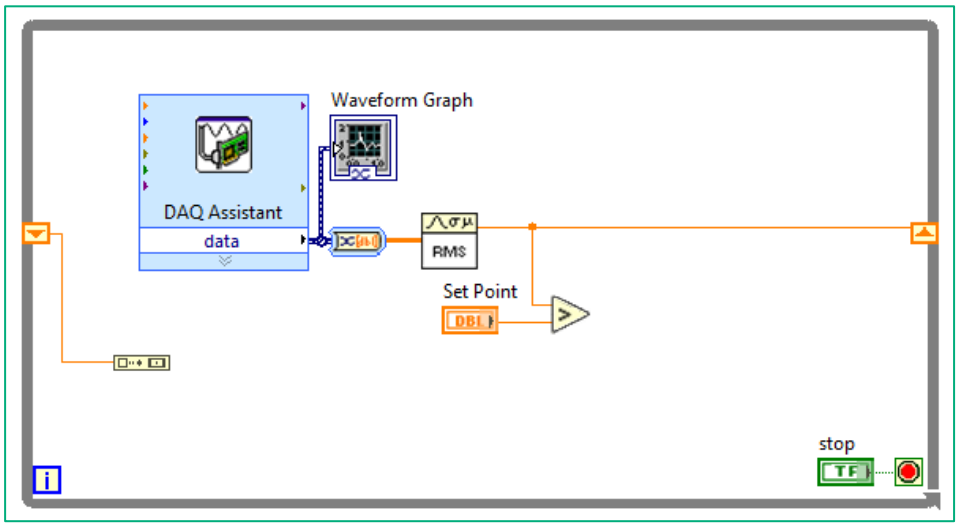


Figure 30. The RMS and set point values are double-precision, floating point data types (DBL) which are denoted by their terminals and wires.

6. To trigger the LED based on the TRUE/FALSE output of the comparison function, add another DAQ Assistant. (Hint: Use the Quick Drop menu again with CTRL+Space). Click inside the while loop to place it.

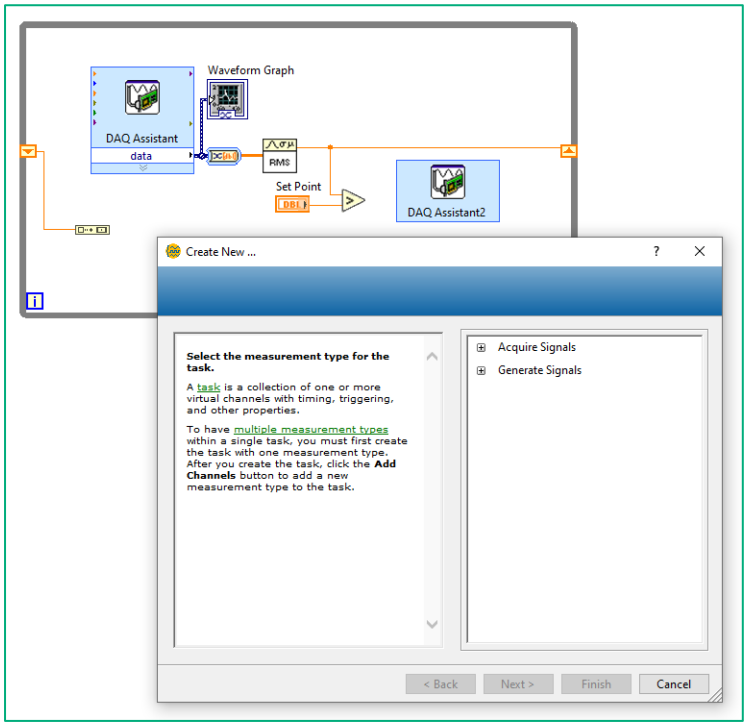


Figure 31. You will use this DAQ Assistant to generate a signal rather than acquire one.

7. To configure the digital signal in the **Create New** window that pops up, expand the **Generate Signals** field, then the **Digital Output** field.

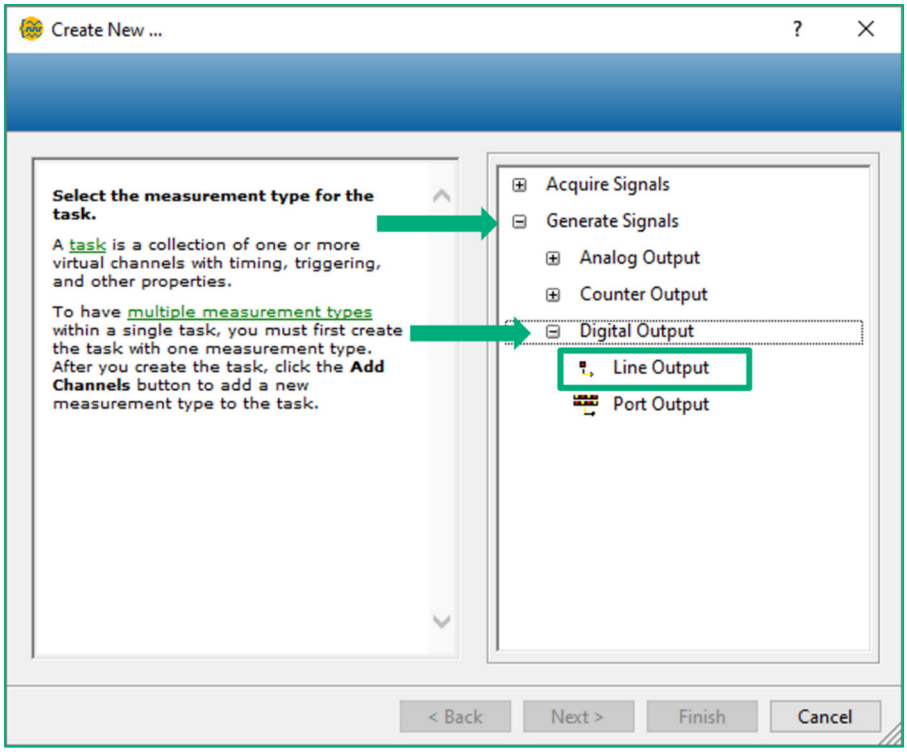


Figure 32. A line is an individual signal, and it refers to a physical terminal.

8. Click **Line Output**

9. Select **port0/line4** as the physical channel and click OK.

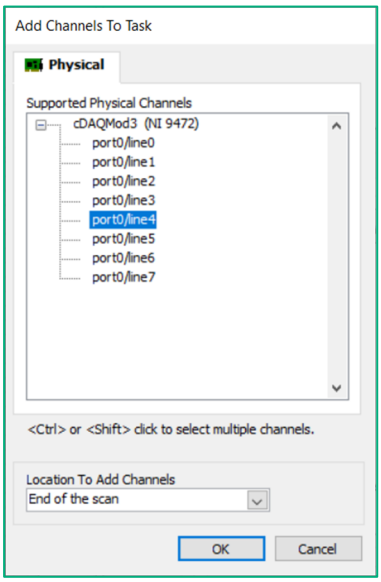


Figure 33. There are eight LEDs on the left side of the demo box, and Port0/line4 corresponds to LED4.

10. In the next window, under the Configuration tab, leave the default settings as shown below, with Generation Mode set to 1 Sample (On Demand).

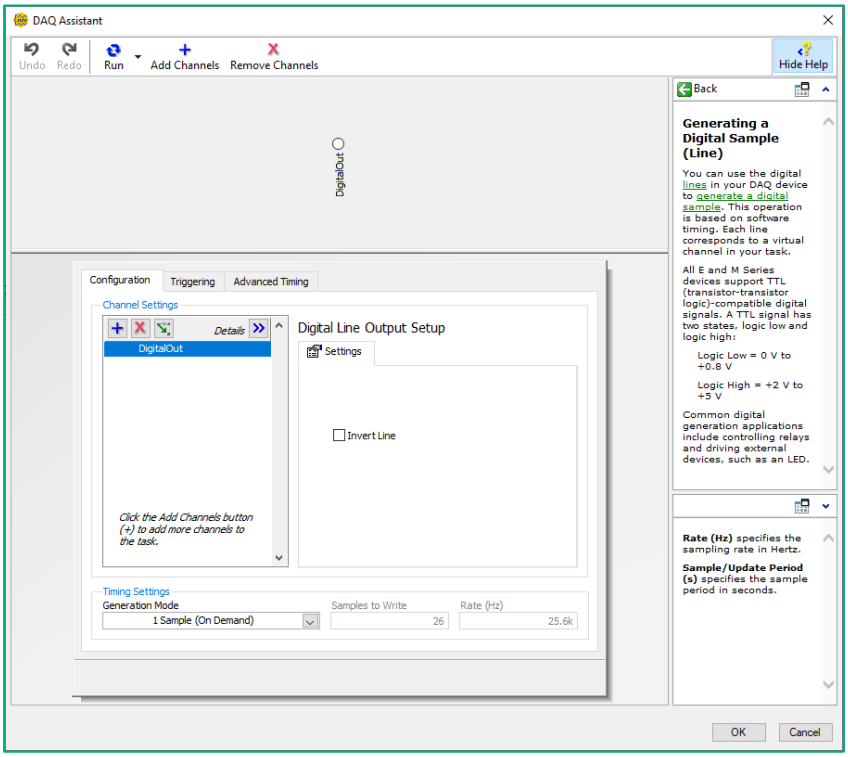


Figure 34. Each loop iteration, the RMS value will be compared with the set point and a signal will be generated to illuminate the LED if the RMS exceeds it.

11. To check that the LED is working, click Run at the top and toggle the **DigitalOut** on and off by clicking it.

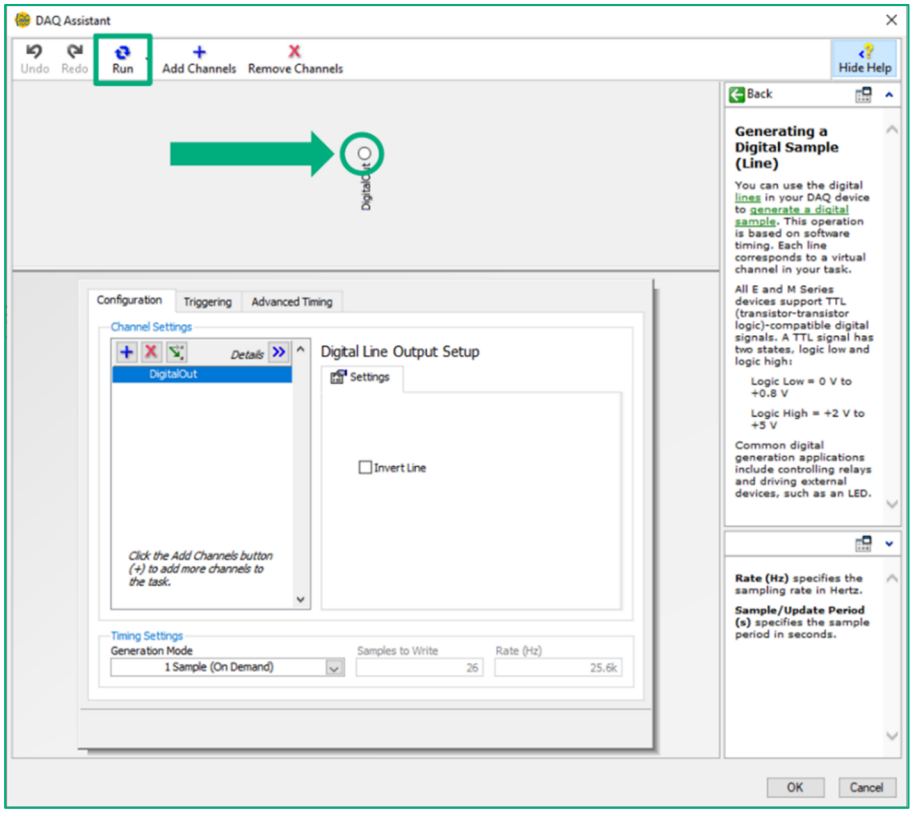


Figure 35. The LED should change state each time you click it.

12. Click **OK** when finished.

13. Access the Quick Drop menu and type the search term “Convert from Dynamic Data”. Double click the option and place it between the comparison function and the DAQ Assistant2.

14. In the window that pops up, under **Scalar Data Type**, click **Boolean (TRUE and FALSE).**

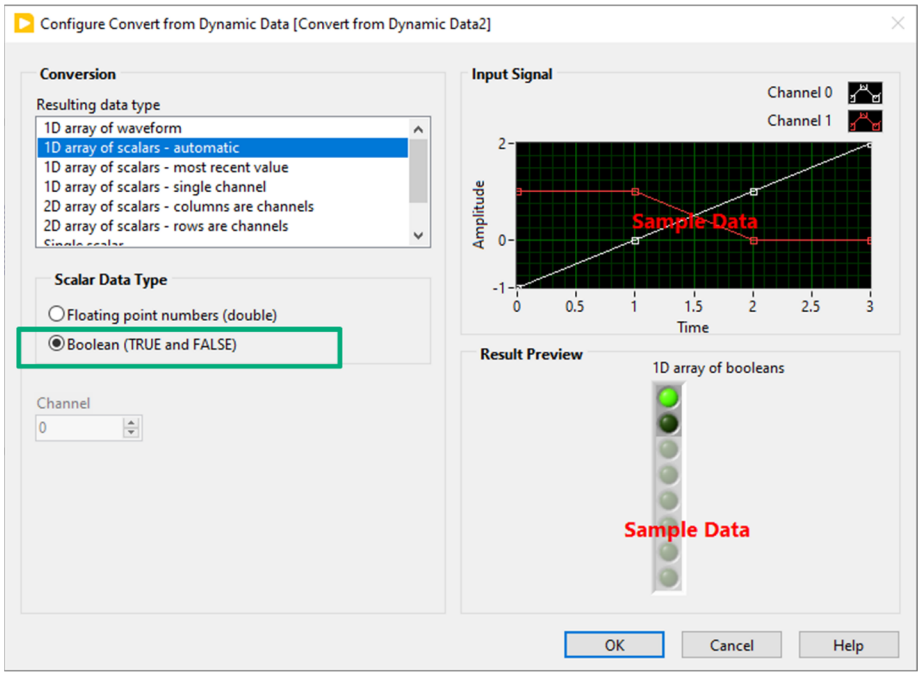


Figure 36. Configuration window for the Convert from Dynamic Data function.

15. Click OK.

16. Next, wire the output of the comparison function to the input of the DAQ Assistant2. You may need to adjust some wires so that the connections are clear.

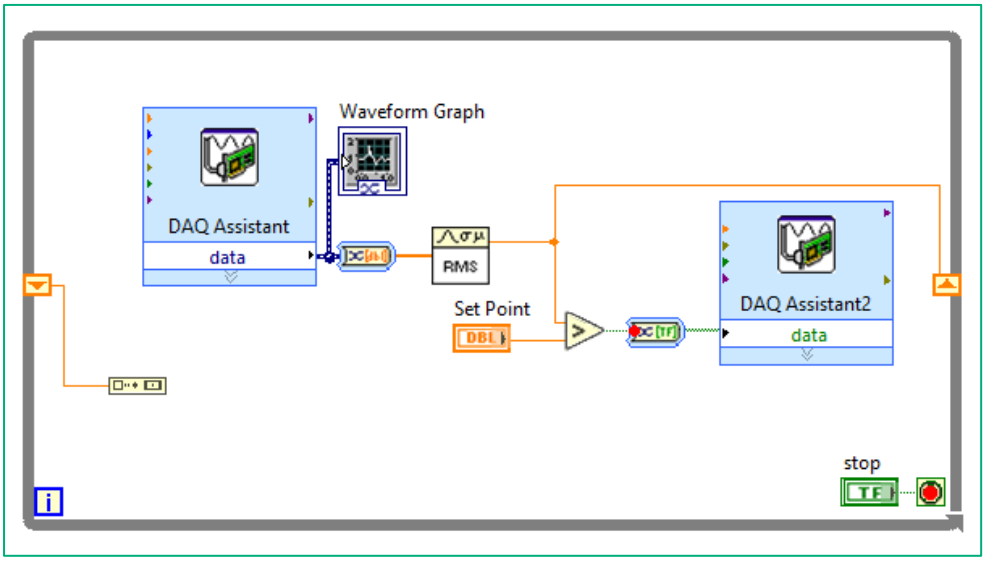


Figure 37. The Block Diagram should now look like this.

17. Navigate to the Front Panel using the shortcut CTRL+E and enter **0.003** for the Set Point.

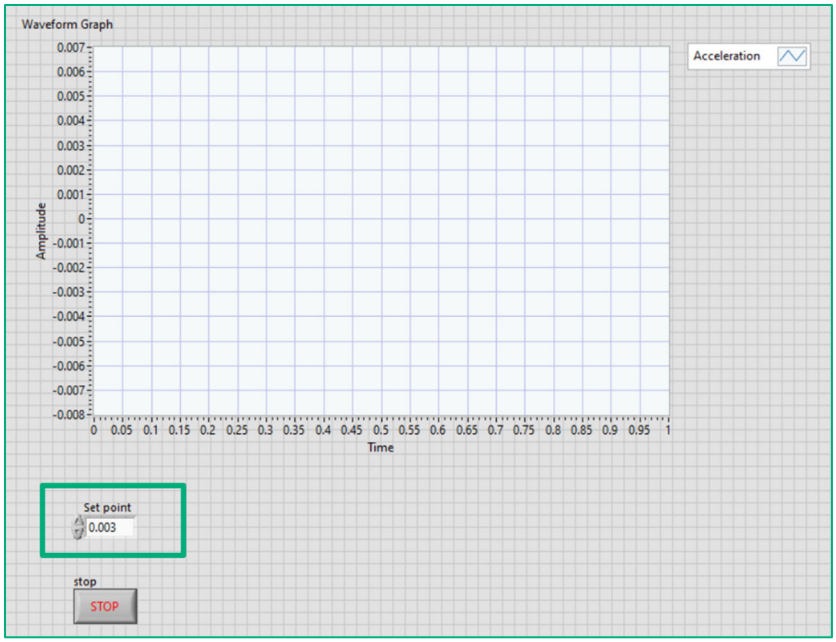


Figure 38. The LED will illuminate when the RMS of the signal exceeds the entered value.

18. Run the VI to start collecting data.

19. Turn the fan speed up or down with the knob and toggle the fan switch back and forth from **Balanced Fan** to **Unbalanced Fan**. Observe the response of the LED, as you change the operating conditions of the fan.

20. When finished, press the STOP button.

Exercise Key Concepts:

In this exercise, you used a second DAQ Assistant to generate a digital output signal, which illuminated an LED whenever an RMS acceleration value exceeded the set point.

<End of Exercise>

Part D: Save the Data to a File

Estimated time: 25 minutes.

In Part D, you will save the acquired data to a TDMS file so you can share it with colleagues, create a report, run additional analysis, or all the above.

1. You must first create a TDMS file to which you will write your data. To do this, use the **TDMS Open Function**.

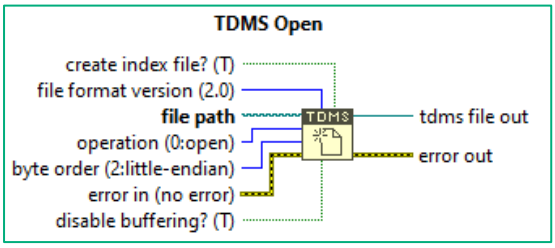


Figure 39. The TDMS Open Function opens a .tdms file for reading or writing, and you also can use this function to create a new file or replace an existing file.

2. Right click the Block Diagram and navigate to **File I/O** > **TDMS Streaming** > **TDMS Open.**

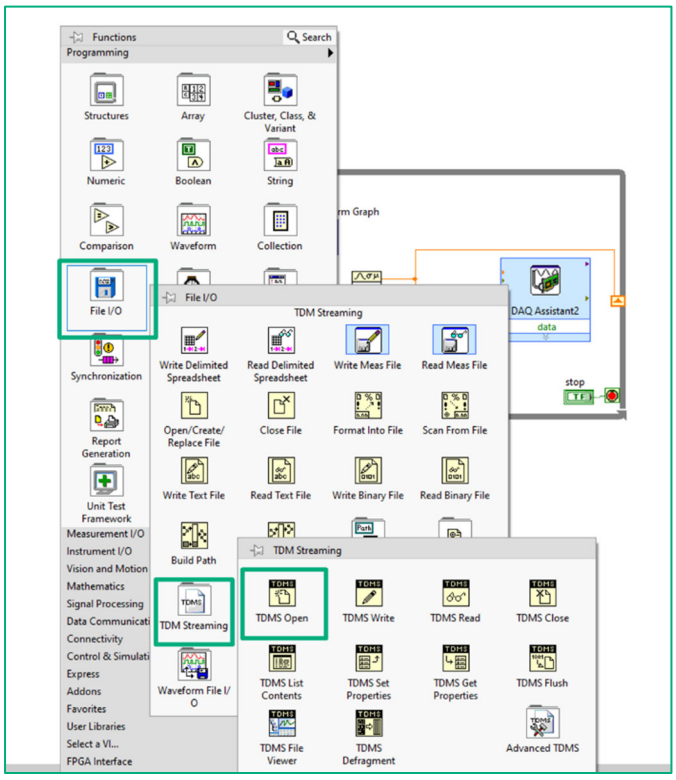


Figure 40. Use the TDM Streaming VIs and functions to read and write waveforms and waveform properties to binary measurement files (.tdms).

3. Click **Open/Create/Replace File** and place it outside the while loop on the left-hand side.

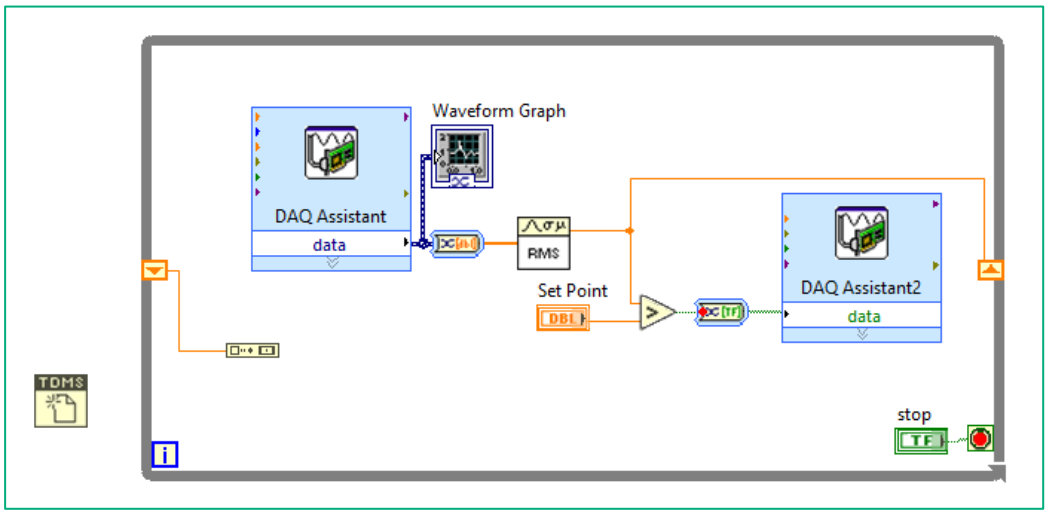


Figure 41. The TDMS function is placed outside the while loop to create a single file to write all data.

4. To set a file path, right click the file path input of the **Open/Create/Replace File** function and click **Create > Control**. Resize the icon by right-clicking and select **View As Icon**.

5. Next, right-click the **Operation** input of the TDMS Open Function and click **Create** > **Constant**. Click the constant to change it to **Create or Replace**.

6. Right-click the **Create Index File?** input and Click **Create** > **Constant**. Leave as the default, which will be a boolean TRUE.

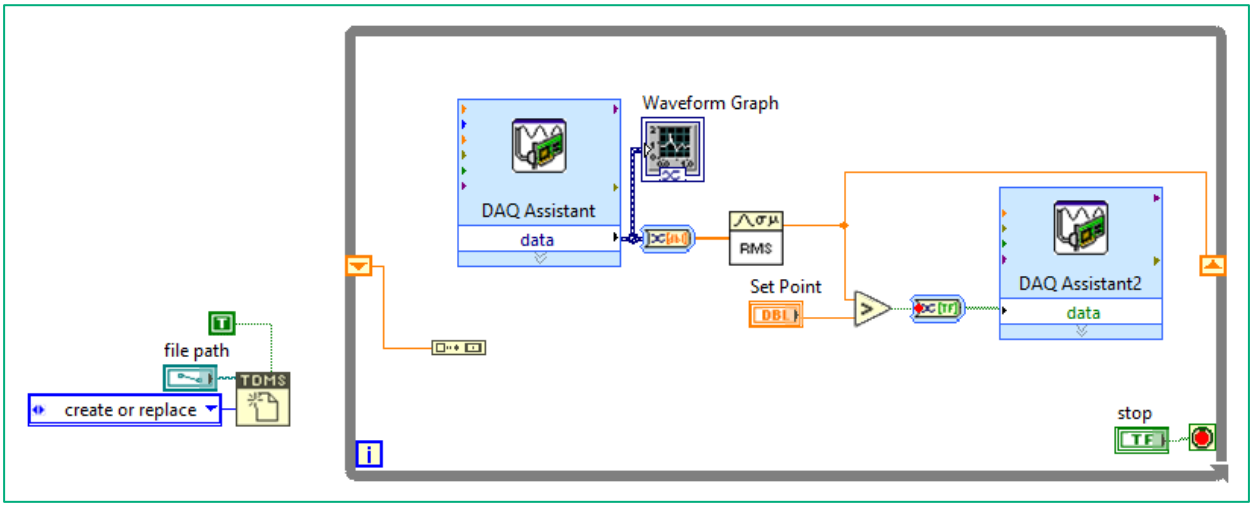


Figure 42. The TDMS Open function creates or replaces an existing file to write data each time the VI runs.

7. Next, use the **TDMS Write Function** to write your data to a file.

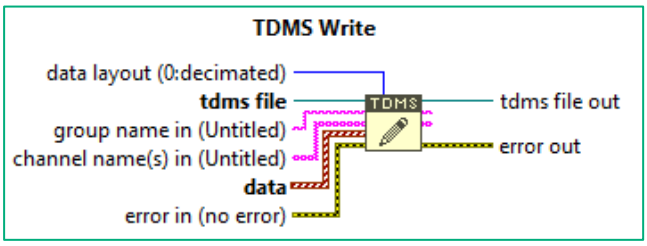


Figure 43. The data subset to write is determined by the values you identify in the group name in and channel name(s) in inputs.

8. Right click the Block Diagram and Navigate to **File I/O** > **TDMS Streaming** > **TDMS Write**. Click **TDMS Write** and click to place the function inside the while loop.

9. Wire the build array data to the **data** input of the **TDMS Write function**.

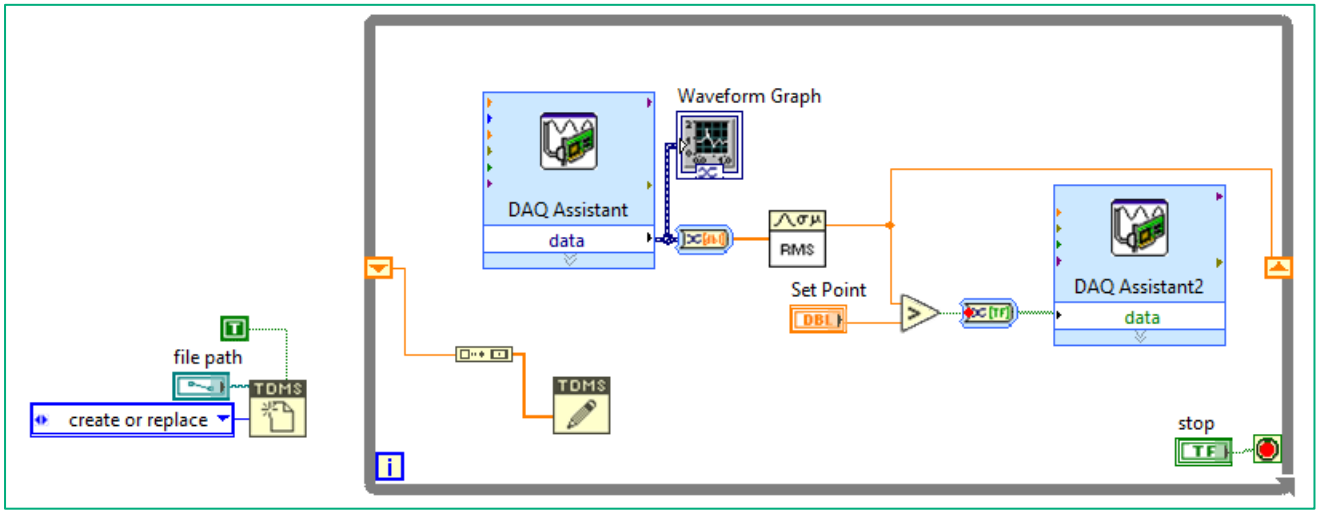


Figure 44. The TDMS Write Function will store the RMS value from each iteration to the same file.

10. To name the channel of data, right click the **channel name(s)** input of the **TDMS Write Function** and click **Create** > **Constant**. Double click in the right cell and type **RMS**.

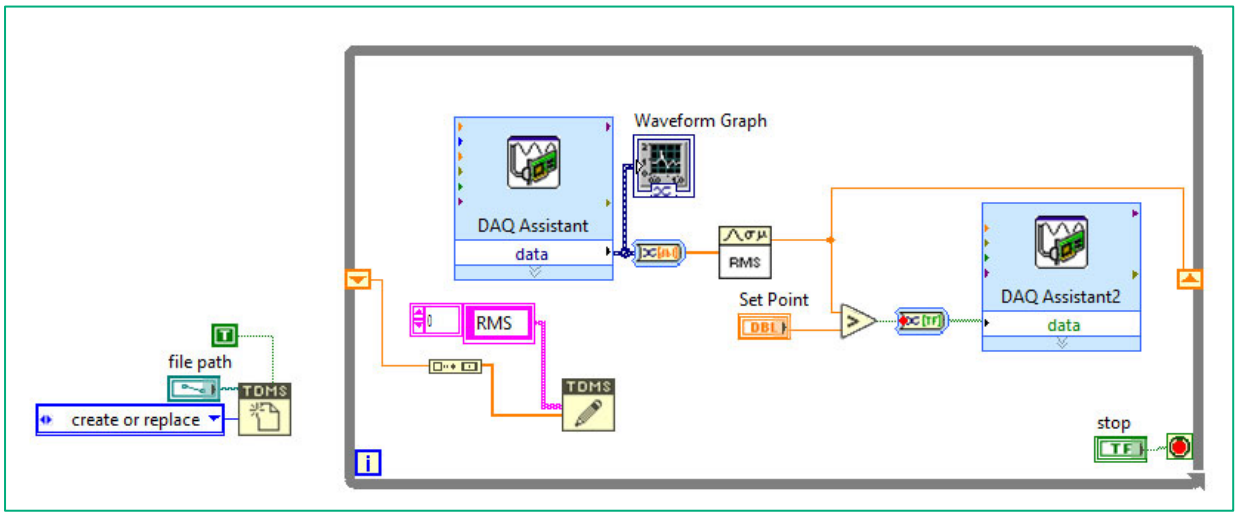


Figure 45. If you do not wire data to channel name(s) in input, LabVIEW names the channels automatically.

11. Add another **TDMS Write Function** to the right of the first one. Wire the raw acceleration data to this one.

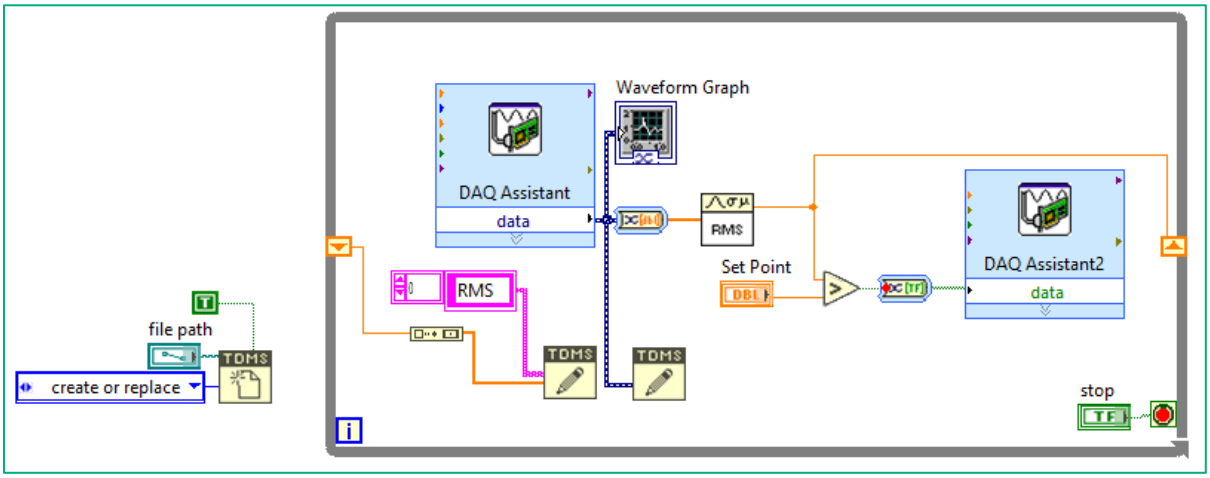


Figure 46. Even though there are two TDMS Write functions, they will write data to the same file.

12. The last function you will add is the **TDMS Close Function.**

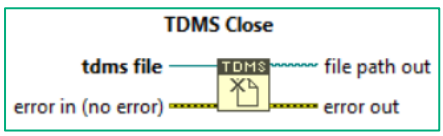


Figure 47. The TDMS Close Function closes the file you opened with the TDMS Open Function.

13. Right click the Block Diagram and Navigate to **File I/O** > **TDMS Streaming** > **TDMS Close**.

14. Click **TDMS Close** and place it outside the while loop, on the right-hand side.

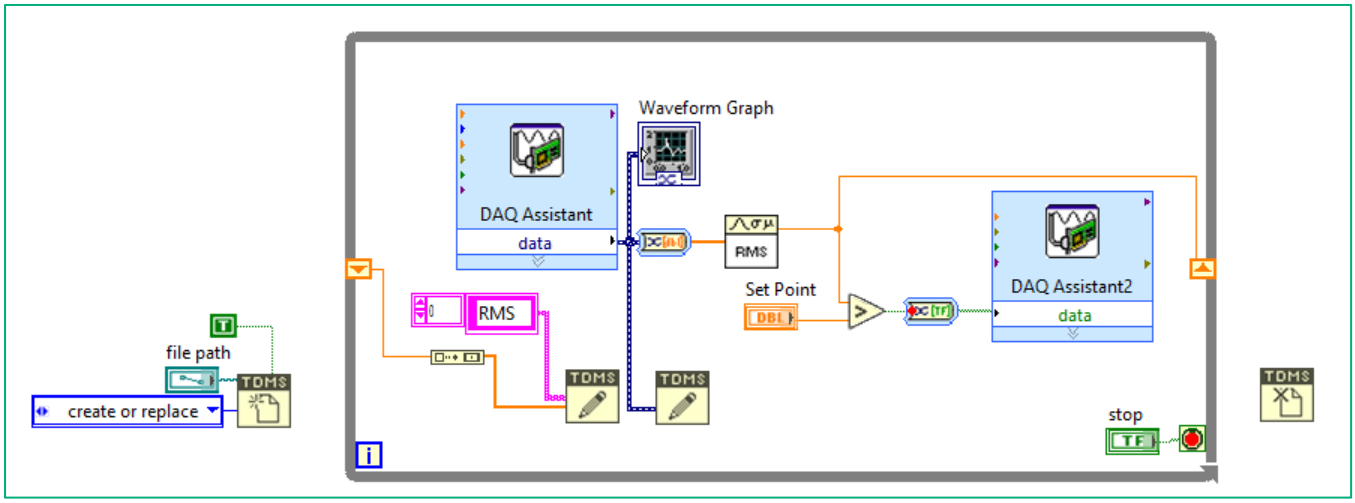


Figure 48. The Block Diagram has all the necessary functions for saving data to a file, but a critical element is missing.

15. Wire the functions together so that the data will be passed between them.

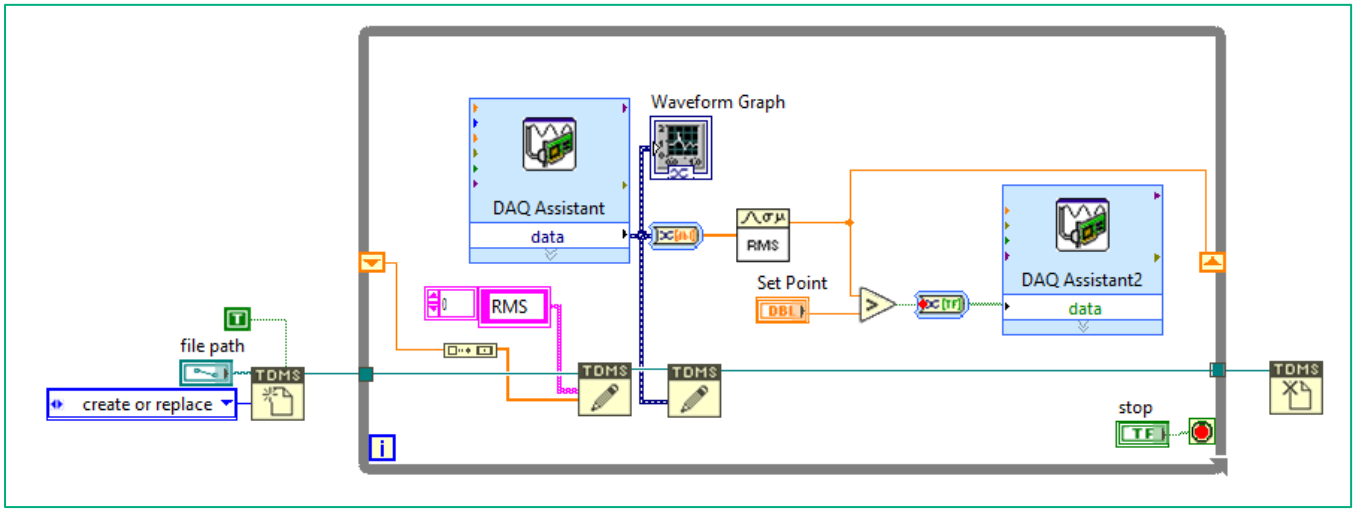


Figure 49. Wire the file paths of the TDMS functions together.

1. Next, add error handling to the application. Start by wiring the bottom output of the **TDMS Open Function** to the first **TDMS Write Function** and continue that pattern of wiring for the remaining functions, ending with the **TDMS Close Function**.

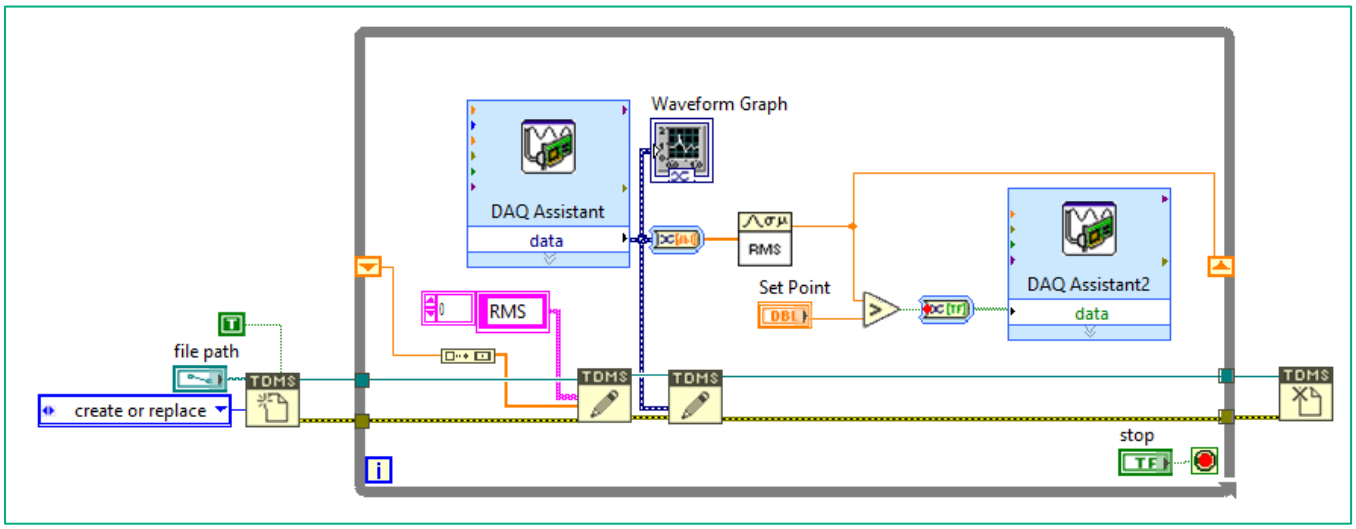


Figure 50. Without a mechanism to check for errors, you can only know that a VI does not work properly.

16. The Block Diagram is now complete.

17. Next, navigate to the Front Panel (Hint: use CTRL+E). In the **file path** field, type **C:\acceleration.tdms**. When you run the VI, LabVIEW will save your vibration data to a file of that name in the given filepath.

18. Next, Run the VI to collect data.

19. Turn the fan speed up or down with the knob and toggle the fan switch back and forth from **Balanced Fan** to **Unbalanced Fan**. Observe the effect of changing the fan conditions on the LED and raw acceleration signal.

20. When finished, press the STOP button.

21. To view your data, open the file explorer on your PC and search **acceleration.tdms**. Right click the .tdms file > **Open With** > **DIAdem**.

Exercise Note:

DIAdem is application software that helps engineers accelerate post-processing of measurement data. With DIAdem, you can quickly automate your workflow, write advanced scripts for complete control, and develop custom applications for others.

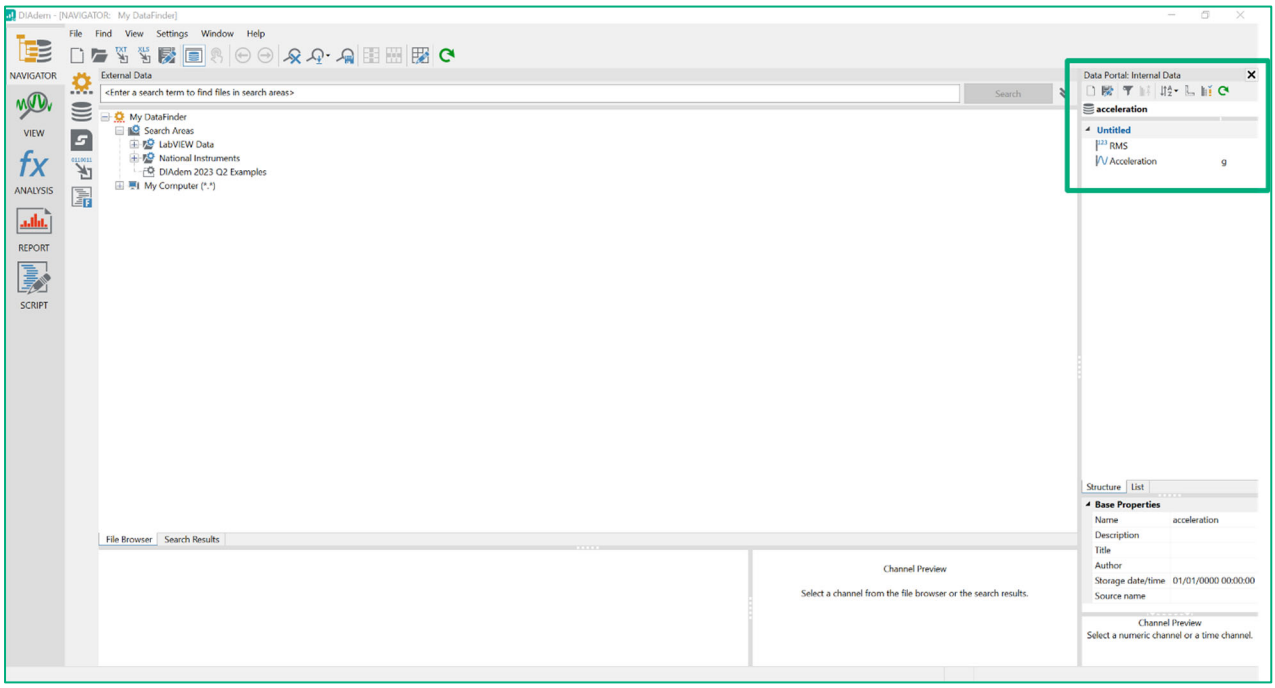


Figure 51. The measurement channels are located in the Data Portal on the right-hand side of the Navigator tab.

22. Click the **View** tab on the left-hand side, underneath the **Navigator** tab.

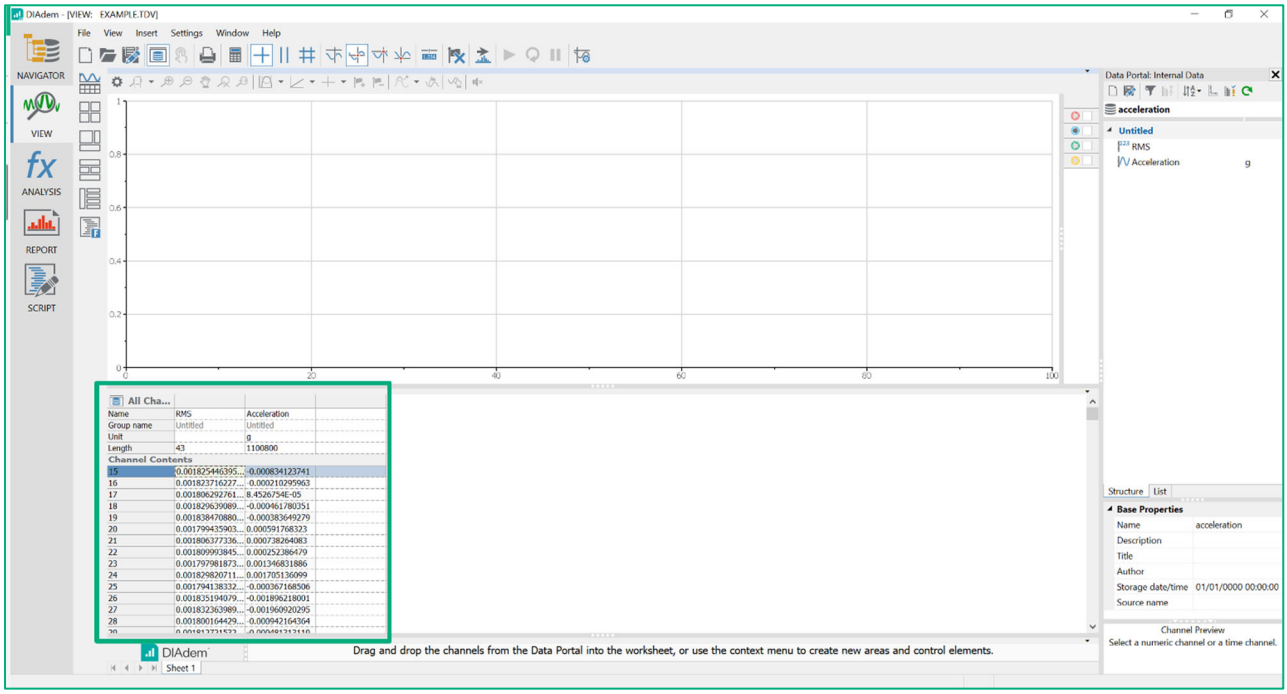


Figure 52. The Channel Contents shows the raw acceleration data as well as the RMS data.

23. Click and drag the **RMS** and **Acceleration** channels from the Data Portal to the graph area.

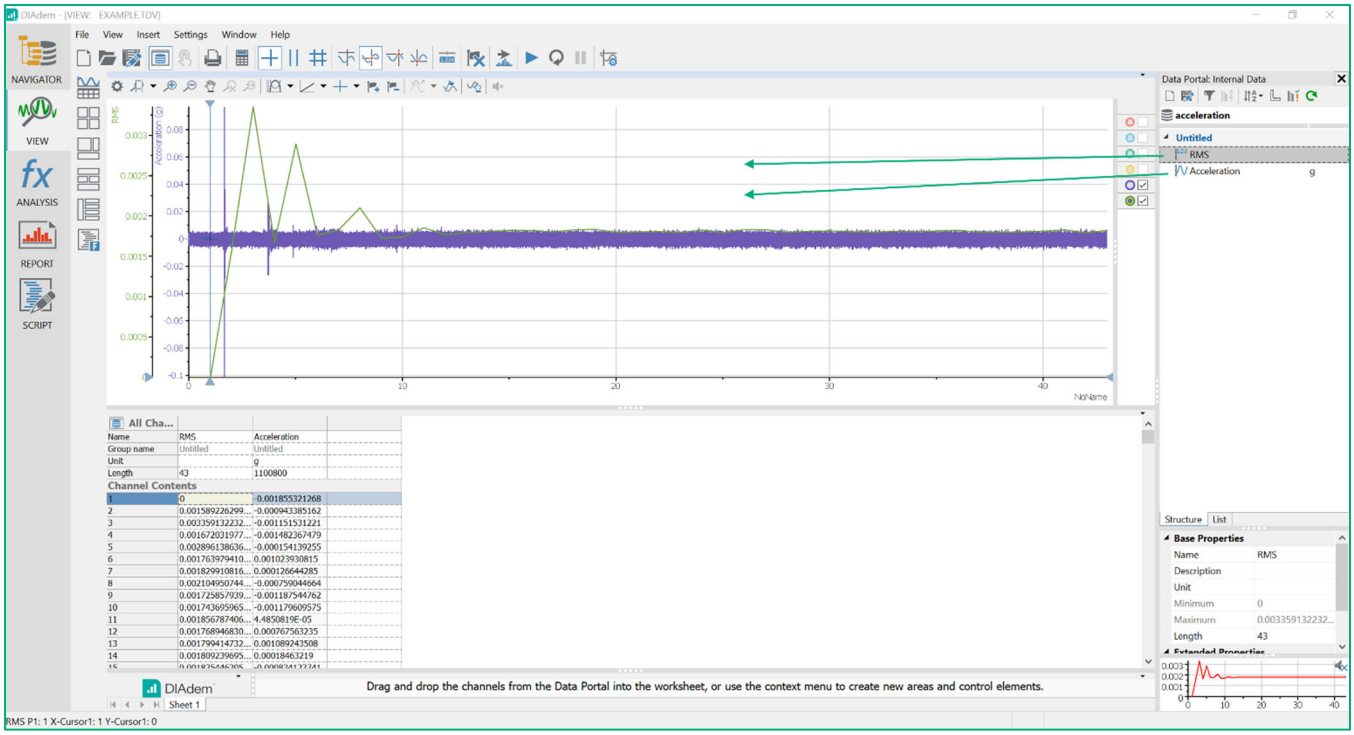


Figure 53. Acceleration and RMS Acceleration are now shown on the same graph.

24. Now you can further analyze the data. Explore the **Analysis**, **Report** and **Script** Tabs on the left pane to become more familiar with DIAdem.

Exercise Key Concepts:

In this exercise, you used the **TDMS Streaming VIs** to save your raw and RMS acceleration data to a TDMS file and opened it in DIAdem for later viewing, analysis, or sharing with a colleague.

<End of Exercise>